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AI-augmented personalized treatment in healthcare a comprehensive survey of models benchmarking and deployment challenges

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Abstract

Personalized Treatment Recommendation Systems (PTRS) are increasingly central to precision medicine, enabling data-driven selection of therapies tailored to individual patient characteristics, including clinical history, biological markers, and behavioral context. Despite rapid methodological advances, existing research remains fragmented, with limited consensus on evaluation practices and insufficient consideration of real-world deployment constraints. This survey provides a systematic and critical review of 60 peer-reviewed studies published between 2019 and 2024, selected using a PRISMA guided protocol from major scholarly databases. The reviewed literature is organized into a unified taxonomy encompassing content-based, collaborative filtering, knowledge-based, and hybrid treatment recommendation paradigms, with emphasis on their clinical applicability and algorithmic design. Recent advances in deep learning and reinforcement learning are examined for their role in improving adaptability and treatment personalization. To enable meaningful cross-study comparison, this work introduces a multidimensional benchmarking rubric that evaluates PTRS beyond predictive performance, incorporating criteria such as algorithmic rigor, clinical relevance, validation depth, data transparency, scalability, and explainability. The analysis reveals persistent challenges related to model interpretability, population generalizability, data privacy, and ethical compliance, which continue to hinder clinical adoption. Emerging solutions, including federated learning, explainable AI, and privacy-preserving architectures, are discussed as promising pathways toward trustworthy deployment. By synthesizing methodological trends, practical limitations, and evaluation gaps, this survey offers a structured foundation and forward-looking perspective for the development of robust, transparent, and clinically viable PTRS in modern healthcare.

Keywords Personalized treatment recommendation systems, Artificial Intelligence in healthcare, Machine learning, Explainable AI

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1 Introduction

The effectiveness of medical treatment is inherently influenced by patient heterogeneity, including biological variability, disease progression patterns, comorbidities, and individual response to therapy. Despite advances in clinical guidelines and evidence-based medicine, treatment selection in real-world healthcare settings remains challenging, particularly for chronic and multifactorial conditions such as diabetes, cardiovascular disorders, cancer, and mental health diseases. Standardized treatment protocols, while clinically validated at population scale, often fail to accommodate patient-specific differences, leading to suboptimal outcomes, adverse drug reactions, and increased healthcare burden. Recent studies report that inappropriate or non-personalized treatment decisions contribute to millions of preventable hospitalizations annually, underscoring the urgent need for individualized therapeutic strategies.

Personalized medicine has emerged as a transformative paradigm to address these limitations by tailoring clinical interventions according to individual patient characteristics, including phenotypic traits, biomarkers, genomic profiles, lifestyle factors, and psychosocial context. Unlike the traditional "one-size-fits-all" approach, personalized treatment strategies aim to optimize therapeutic efficacy while minimizing side effects and unnecessary interventions. Empirical evidence suggests that personalized treatment approaches can improve therapeutic effectiveness by up to 30% and significantly reduce adverse reactions, particularly in chronic disease management. The growing clinical and technological maturity of this paradigm has led to a rapid increase in the adoption of personalized treatment strategies over the past decade, driven by advances in artificial intelligence (AI), large-scale health data analytics, and digital health infrastructure.

1.1 Background and motivation

Historically, clinical decision-making has relied on generalized models derived from controlled clinical trials and consensus-driven guidelines. While such approaches provide a strong foundation for evidence-based care, they often underrepresent patients with multimorbidity, rare conditions, or atypical disease trajectories. As a result, clinicians frequently face uncertainty when translating guideline recommendations into individualized treatment plans. This gap is further exacerbated by health disparities across demographic and socioeconomic groups, where generalized treatments may not align with patient-specific risks or access constraints.

Personalized Treatment Recommendation Systems (PTRS) aim to bridge this gap by systematically integrating heterogeneous patient data and computational intelligence to support individualized treatment selection. PTRS leverage a combination of electronic health records (EHRs), clinical measurements, genomics, behavioral data, and historical treatment outcomes to generate patient-specific therapeutic recommendations. The increasing availability of high-dimensional medical data, coupled with advances in machine learning, deep learning, and reinforcement learning, has significantly accelerated research in this area. However, translating these algorithmic advances into clinically reliable and trustworthy decision-support tools remains an open challenge.

In contemporary healthcare discourse, the terms *personalized medicine* and *precision medicine* are often used interchangeably, although they reflect distinct conceptual emphases. Precision medicine primarily focuses on stratifying patient populations using molecular, genetic, and biomarker-driven information to identify subgroup-specific

treatment responses. In contrast, personalized medicine adopts a broader perspective by integrating not only biological and clinical attributes, but also behavioral, environmental, and contextual factors to tailor care at the individual patient level. In this survey, the term *personalized treatment recommendation systems (PTRS)* is used as an overarching concept encompassing algorithmic frameworks that operationalize both precision-driven stratification and individualized treatment adaptation. When the term *precision medicine* appears, it reflects the terminology adopted in the cited studies or contexts explicitly centered on genomics or biomarker-based decision-making.

1.2 Overview of recommendation techniques

AI-driven recommender systems constitute the computational backbone of modern PTRS. These systems adapt methodologies originally developed for information retrieval and decision support to the clinical domain, where correctness, transparency, and safety are paramount. As illustrated in Fig. 1, existing treatment recommendation approaches can be broadly categorized into four major paradigms.

Content-Based Filtering (CBF) focuses on matching patient profiles with treatment characteristics by analyzing clinical features such as symptoms, laboratory results, and historical responses. Collaborative Filtering (CF) identifies patterns across patient populations by exploiting similarities in treatment outcomes among clinically comparable patients. Knowledge-Based Filtering (KBF) relies on structured medical knowledge, clinical guidelines, and expert-defined rules, making it particularly valuable when patient interaction data are sparse or unavailable. Hybrid Recommendation Systems combine multiple paradigms to mitigate the limitations of individual methods, addressing challenges such as cold-start conditions, data sparsity, and model bias. By integrating data-driven learning with expert medical knowledge, these systems aim to deliver clinically meaningful, personalized, and adaptive treatment recommendations.

1.3 Role of recommender systems in personalized treatment

Recommender systems play an increasingly central role in supporting clinical decision-making by transforming complex, multidimensional patient data into actionable

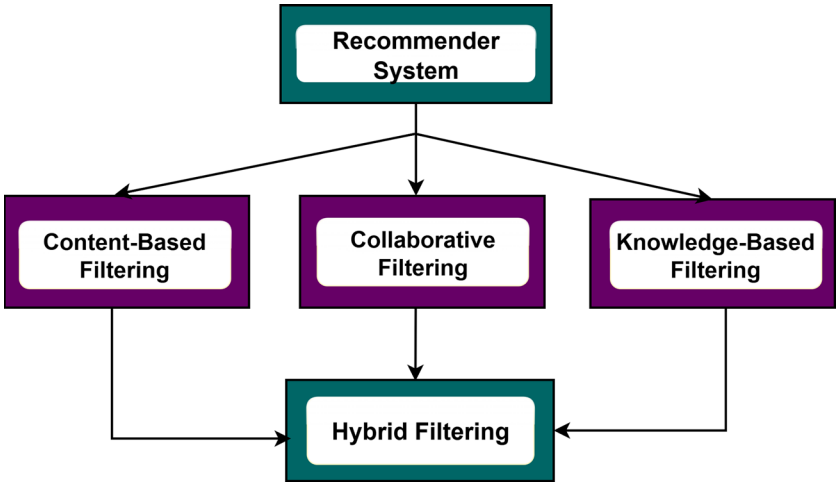


Fig. 1 Key categories of recommendation techniques applied in personalized treatment systems, including Content-Based, Collaborative, Knowledge-Based filtering, and Hybrid models that integrate multiple strategies for improved clinical decision support

treatment insights. In healthcare, unlike consumer recommendation platforms, treatment recommendations directly influence patient safety and clinical outcomes. Consequently, transparency, interpretability, and trust are critical requirements. Black-box models that lack explainability may hinder clinician adoption, even when predictive performance is high. Effective PTRS therefore emphasize explainable AI (XAI) mechanisms, enabling clinicians to understand the rationale behind recommendations and to integrate algorithmic insights with clinical judgment.

Beyond pharmacological decision support, PTRS also facilitate personalized non-pharmacological interventions, including lifestyle modification, behavioral therapy, and long-term disease management strategies. By continuously learning from patient feedback and longitudinal outcomes, these systems support adaptive and patient-centered care pathways. A system-level overview of a personalized treatment recommendation framework is illustrated in Fig. 2, highlighting the interaction between patient data acquisition, feature extraction, recommendation engines, and clinical decision support.

1.4 Gaps in existing surveys and research motivation

Although several survey articles have explored AI-based clinical decision support and healthcare recommender systems, most existing reviews focus on algorithmic development in isolation or emphasize diagnostic applications rather than treatment selection. Many surveys predate recent advances in deep learning, reinforcement learning, federated learning, and explainable AI, or fail to systematically evaluate deployment challenges such as privacy preservation, regulatory compliance, interpretability, and real-world scalability. Furthermore, there is a lack of unified benchmarking frameworks

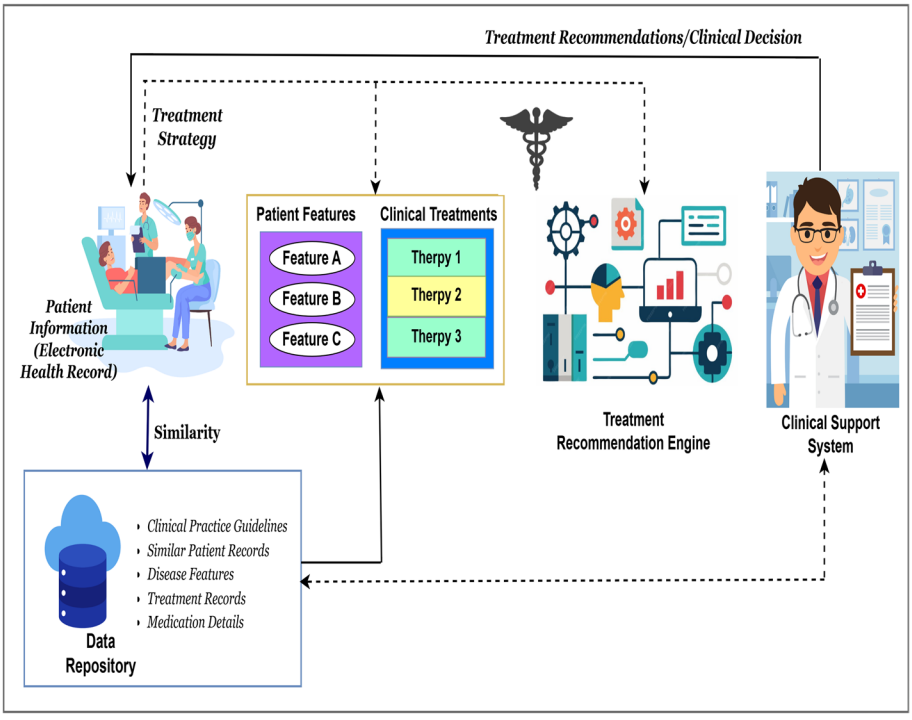


Fig. 2 System-level workflow of a personalized treatment recommendation framework, illustrating how patient data from electronic health records are processed through recommendation models to generate evidence-based and individualized treatment options

that allow meaningful comparison across heterogeneous datasets, evaluation metrics, and clinical contexts.

1.5 Contributions of this survey

To address these limitations, this study makes the following key contributions:

- Provides a comprehensive and up-to-date systematic review of Personalized Treatment Recommendation Systems published between 2019 and 2024, following a PRISMA-guided methodology.
- Develops a structured taxonomy of treatment recommendation paradigms, including content-based, collaborative, knowledge-based, and hybrid approaches, with in-depth technical analysis.
- Introduces a multi-criteria benchmarking rubric that evaluates PTRS models beyond predictive accuracy, incorporating clinical relevance, interpretability, data transparency, and validation rigor.
- Analyzes real-world deployment challenges, including data privacy, security, ethical concerns, and regulatory constraints, alongside emerging mitigation strategies.
- Identifies open research gaps and future directions to guide the development of trustworthy, scalable, and patient-centric treatment recommendation systems.

1.6 Related work and positioning of this survey

Several prior reviews have addressed adjacent areas such as healthcare recommender systems, medication recommendation from EHRs, and AI-enabled clinical decision support. However, these reviews typically (i) treat "health recommendation" broadly (often spanning lifestyle, wellness, and general guidance), (ii) focus on a single technical family (e.g., deep learning for medication recommendation or reinforcement learning for sequential decision-making), or (iii) summarize performance narratively without providing an auditable extraction of datasets, validation designs, and metric types at the study level. As a result, the deployment-facing question that motivates this work *what constitutes credible evidence for personalized treatment recommendation (PTR) systems under heterogeneous clinical tasks and evaluation protocols* remains insufficiently synthesized in a single, corpus-grounded framework.

Our survey is differentiated in three ways. First, we scope specifically to *personalized treatment recommendation* (PTR) tasks (therapeutic choice, regimen/pathway recommendation, or treatment planning) and explicitly separate these from purely diagnostic or risk-prediction studies unless the output is a treatment recommendation. Second, we elevate *evaluation comparability* as a first-class concern by providing a systematic extraction table that records, for each included study, the recommendation paradigm, clinical domain, dataset characteristics, validation approach, and the reported performance indicator(s) (Tables 4, 5, 6 and 7). Third, because reported metrics are often heterogeneous and not meta-analyzable, we add an auditable *evidence-strength* synthesis that captures methodological and translational aspects beyond headline performance (Appendix A, Table 9), complemented with worked scoring examples that document the evidence used to support rubric assignments (Appendix A, Table 10). Together, these contributions position the manuscript as a survey that supports *verifiable cross-study synthesis* rather than narrative comparison alone, while keeping claims bounded by what

the corpus reports. A concise comparison with representative prior reviews is provided in Table 1.

1.7 Scope and organization of the paper

This survey focuses exclusively on AI-driven treatment recommendation systems in healthcare, excluding diagnostic only models and non-clinical recommender applications. The review encompasses peer-reviewed journal and conference publications written in English and published between 2019 and 2024.

The remainder of this paper is structured as follows. Section 2 details the systematic review protocol, including the formulation of research questions, database selection, search strategy, eligibility criteria, and study screening process. Section 3 presents a synthesis of key findings, offering cross-domain insights into dominant recommendation paradigms, disease-specific applications, algorithmic trends, and benchmarking outcomes. Section 4 critically examines translational challenges associated with real-world deployment, including interpretability, privacy and security risks, fairness considerations, and regulatory constraints, alongside emerging mitigation strategies. Finally, Sect. 5 concludes the survey by summarizing the principal insights and outlining future research directions toward transparent, scalable, and clinically actionable personalized treatment recommendation systems.

For readability, Table 2 lists the abbreviations used throughout the manuscript.

2 Systematic review methodology and key insights

This section describes the systematic methodology adopted to ensure a comprehensive, transparent, and unbiased synthesis of the literature on personalized treatment recommendation systems (PTRS). In line with established guidelines for systematic reviews, the study was designed to address the following research questions, which collectively guide the scope of analysis and subsequent benchmarking.

2.1 Research questions

The review is structured around four key research questions:

Table 1 Positioning relative to representative prior reviews on healthcare recommendation and AI-enabled decision support

Prior review	Primary scope	Typical tasks	Synthesis emphasis	Gap addressed here
[1]	Health recommender systems (broad)	Lifestyle and wellness guidance; mixed medical contexts	Systematic categorization with limited PTR-specific evaluation comparability	We focus on PTR tasks and provide study-level extraction of datasets, validation strategies, and metrics (Tables 4, 5, 6 and 7).
[2]	Medication recommendation from EHRs	Medication and drug recommendation using prediction-driven pipelines	Model families and EHR-centric methods with limited deployment constraints across PTR tasks	We broaden beyond medication-only studies to PTR modalities and add evidence-strength synthesis (Tables 9, 10).
[3]	Reinforcement learning in healthcare	Sequential decision-making and policy learning for treatment	Algorithmic formulation and RL evaluation patterns	We integrate RL and CF within a paradigm-level PTR taxonomy and compare against other paradigms using auditable extraction (Tables 4, 5, 6 and 7).

The table highlights scope and synthesis granularity; it is illustrative rather than exhaustive

Table 2 List of abbreviations used in this study

Abbreviation	Full form
PTRS	Personalized treatment recommendation systems
CBF	Content-based filtering
CF	Collaborative filtering
MLP	Multilayer perceptron
CNN	Convolutional neural networks
AI	Artificial Intelligence
ML	Machine learning
SVM	Support vector machines
KNN	k-Nearest neighbors
RL	Reinforcement learning
DQN	Deep Q-Network
SAI-DQN	Clinical knowledge-guided deep Q-Network
EMA	Ecological momentary assessment
EHR	Electronic health records
NN	Neural network
ARIC	Atherosclerosis risk in communities
TCGA	The cancer genome atlas
REVERT	A project database for oncology patients
XAI	Explainable AI
LSTM	Long short-term memory
BiLSTM	Bidirectional long short-term memory
GRU	Gated recurrent unit
CDSS	Clinical decision support system
DNA	Deoxyribonucleic acid
EEG	Electroencephalogram
TREC	Text retrieval conference
FMRI	Functional magnetic resonance imaging
ROC	Receiver operating characteristic
KBF	Knowledge-based filtering

- *RQ1.* What types of recommendation algorithms and techniques are employed in personalized treatment recommendation systems (PTRS)?
- *RQ2.* Which datasets are utilized in PTRS studies for the experimental validation of treatment recommendation techniques?
- *RQ3.* What key findings and limitations are reported in existing PTRS studies?
- *RQ4.* What emerging research trends and future directions can be identified in PTRS research?

These research questions are designed to capture both methodological and practical dimensions of PTRS, enabling a structured analysis of algorithmic approaches, data sources, empirical evidence, and open research challenges.

2.2 Literature search strategy

A comprehensive and systematic literature search was conducted to identify relevant studies on AI-enabled personalized treatment recommendation systems. Major scholarly repositories spanning computer science and medical research were explored, including PubMed, IEEE Xplore, ScienceDirect, SpringerLink, Web of Science, ProQuest, Scopus, and Google Scholar. These databases were selected due to their broad coverage of peer-reviewed research in artificial intelligence, healthcare informatics, and clinical decision support systems.

To retrieve a focused yet inclusive set of studies, structured search queries were formulated by combining key concepts related to personalized treatment (e.g., "personalized treatment", "individualized care") with artificial intelligence paradigms (e.g., "artificial intelligence", "machine learning", "deep learning") and healthcare-specific contexts (e.g., "healthcare", "clinical decision support"). Boolean operators were employed to refine the search and reduce irrelevant retrievals. This strategy ensured coverage of recent advancements while maintaining thematic relevance to PTRS.

The search was restricted to studies published between 2019 and 2024 to capture contemporary developments in AI-driven treatment recommendation. Only full-text articles published in peer-reviewed journals and conference proceedings were considered. Publications written in English were included, while books, dissertations, editorials, patents, white papers, surveys, and other forms of gray literature were excluded to maintain methodological rigor and consistency.

Each retrieved article was manually screened for relevance to personalized treatment recommendation in the medical domain. Studies that employed recommendation techniques in non-clinical contexts such as cosmetic or consumer-oriented recommendations were excluded to preserve the clinical focus of the review. Eligible studies were required to provide detailed descriptions of recommendation algorithms, system architectures, learning models, or evaluation frameworks, along with experimental validation using real or simulated medical data.

To support qualitative analysis of research trends, a keyword co-occurrence network based on titles and abstracts was constructed, as illustrated in Fig. 3. This visualization highlights dominant research themes and methodological clusters within the PTRS literature. Furthermore, the included studies were required to report empirical outcomes demonstrating the effectiveness of personalized treatment recommendations. The inclusion and exclusion criteria applied during the selection process are summarized in Table 3.

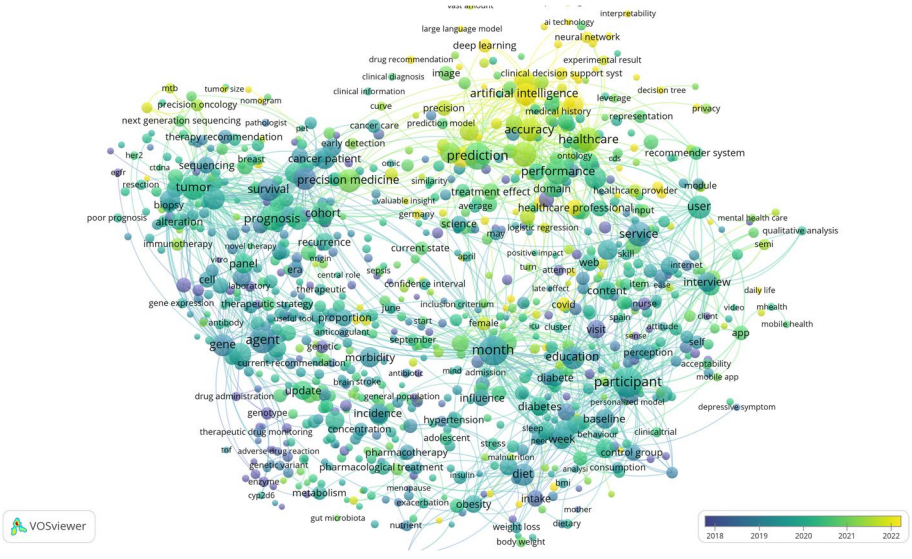


Fig. 3 Keyword co-occurrence network of personalized treatment recommendation systems. The visualization emphasizes dominant thematic clusters by increasing node spacing and restricting labels to high-frequency keywords, improving interpretability while preserving the underlying bibliometric structure

Table 3 Inclusion and exclusion criteria for article selection

Inclusion criteria	Exclusion criteria
Articles published between 2019 and 2024	Articles published before 2019
Articles written in English	Non-English language articles
Articles published in reputed journals or conference proceedings indexed in scientific databases such as IEEE, Springer, ScienceDirect, and PubMed	Grey literature and non-primary sources, including books, book chapters, keynotes, posters, panels, discussions, abstracts, courses, encyclopedias, editorials, reports, correspondence, handbooks, and theses
Studies focused on the design, development, or evaluation of a recommendation system for personalized treatment	Review papers, including narrative reviews, systematic reviews, and surveys
Articles that present recommendation algorithms, system architecture, learning model or algorithm design, and related experimental evidence	Articles that use the term "recommendation" in a context unrelated to healthcare or clinical decision-making
Full-text availability and sufficient methodological reporting to support extraction and synthesis	Studies with insufficient methodological details to support extraction *(operationalized as missing one or more of: a clearly specified recommendation method, an identifiable dataset/cohort description, and an explicit evaluation protocol with at least one reported outcome/performance indicator)*
	Duplicate or near-duplicate articles presenting the same methodology with minimal variation across different datasets
	Articles with inaccessible full texts or paywalls despite institutional access

The criteria were carefully defined to minimize bias and enhance reproducibility. Peer-reviewed empirical studies were prioritized for their evidentiary strength, while duplicate or incomplete studies were excluded to avoid analytical distortion. Collectively, this systematic search strategy ensures that the reviewed literature provides a robust and up-to-date foundation for subsequent comparative analysis and benchmarking.

2.3 Analysis of search results

Our database search retrieved 3,230 records from the selected scholarly repositories. During the initial screening stage, we applied exact keyword matching using combinations of *"personalized," "treatment," "recommendation,"* and *"recommender"* connected via Boolean operators (AND/OR), yielding 1,227 records aligned with the search intent.

We then applied predefined eligibility criteria (Table 3) in a stepwise manner. First, 674 records were excluded for duplication, language, and venue-related reasons (e.g., duplicates, non-English papers, and items not published in peer-reviewed journal or conference venues). Next, 394 records were removed for falling outside the 2019–2024 publication window. We further excluded 155 records that were review articles or that discussed "recommendation" in non-treatment or non-clinical contexts. Finally, 65 records were excluded at the abstract/full-text feasibility stage due to restricted access, incomplete availability, or insufficient relevance to treatment recommendation after abstract screening. To reduce selection bias, we attempted access via institutional subscriptions and publisher/author repositories; records were classified as "restricted/incomplete" only when full text could not be obtained or when the abstract/full text did not describe a treatment recommendation task. The complete selection pathway is summarized in Fig. 4.

To improve transparency in borderline decisions, we operationalized the exclusion criterion *"insufficient methodological details"* as missing one or more of: (i) a clearly specified recommendation method, (ii) an identifiable dataset/cohort description, and (iii)

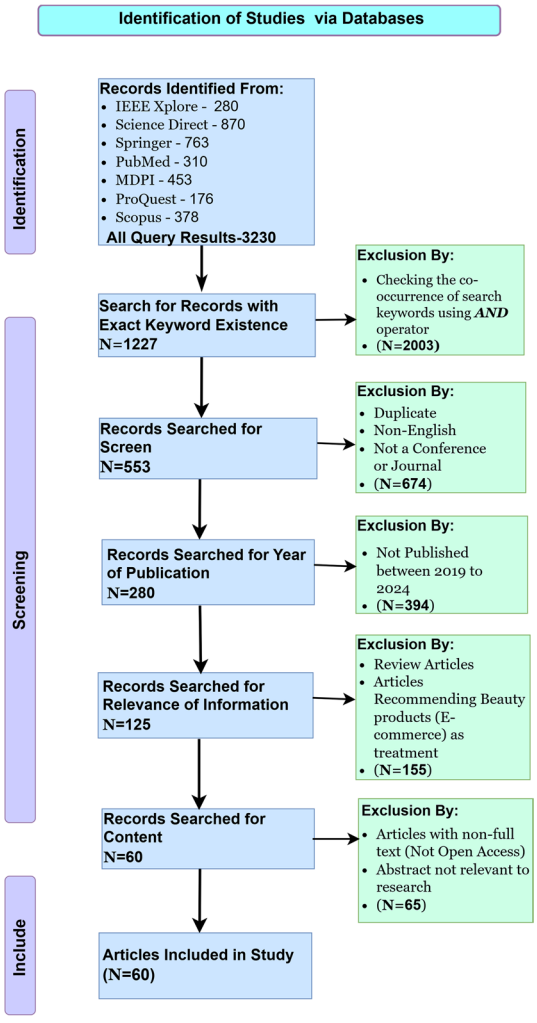


Fig. 4 PRISMA-based flow diagram illustrating identification, screening, eligibility assessment, and final inclusion of studies. A total of 3,230 records were retrieved, and 60 studies were retained after sequential filtering and eligibility checks

an explicit evaluation protocol with at least one reported outcome/performance indicator. For boundary cases between treatment recommendation and adjacent tasks (e.g., diagnosis-only systems), studies were retained only when the primary output explicitly supported a *treatment choice* (e.g., therapy/drug/intervention selection, dosing, care pathway recommendation) rather than solely predicting disease state or risk.

Among the 60 included studies, 44 were journal articles and 16 were conference papers. Figure 5 shows the year-wise distribution from 2019 to 2024. Overall, publication volume increases over time, with journals consistently contributing the larger share, indicating maturation of PTR research into more archival venues.

Figure 6 summarizes the database-wise distribution of the included studies. IEEE Xplore accounts for the largest share of retrieved publications, followed by Springer. The concentration of included studies in these repositories reflects where PTR work is most frequently disseminated, while the rise around 2023 suggests accelerating research activity aligned with broader uptake of AI-enabled clinical decision support.

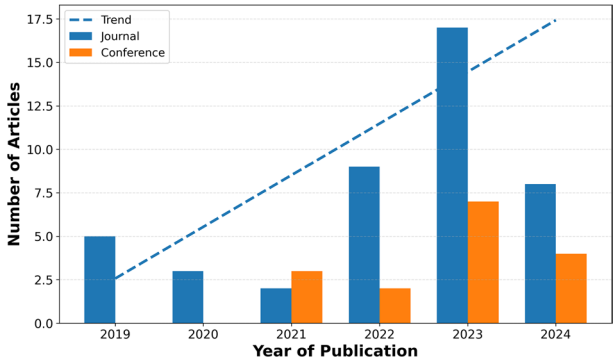


Fig. 5 Year-wise distribution of journal and conference publications (2019–2024), showing publication volume trends over time

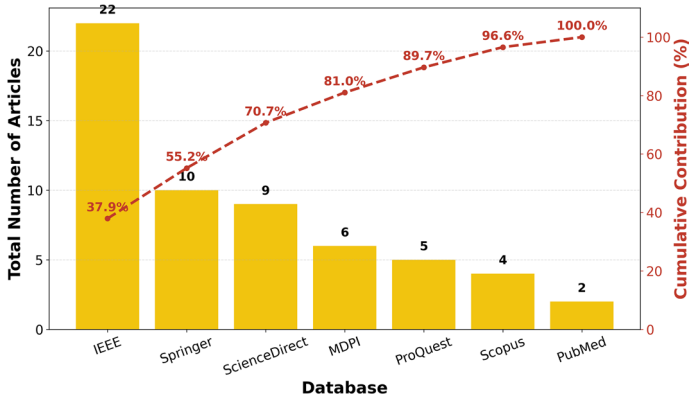


Fig. 6 Database-wise distribution of the included studies (2019–2024) shown as a Pareto-style plot (counts and cumulative percentage)

2.4 Key insights and analysis of treatment recommendation techniques

This subsection presents a structured analysis of the principal algorithmic paradigms employed in personalized treatment recommendation systems (PTRS), focusing on how different recommendation strategies leverage patient-specific data to support individualized clinical decision-making. Personalized treatment recommendation aims to tailor healthcare interventions by accounting for patient characteristics, clinical histories, and contextual constraints, thereby enhancing therapeutic effectiveness and patient outcomes.

To facilitate systematic comparison, the reviewed studies are categorized into content-based filtering (CBF), collaborative filtering (CF), knowledge-based filtering (KBF), and hybrid recommendation approaches. This categorization enables a comparative assessment of each paradigm by highlighting their respective strengths, limitations, and suitability across diverse healthcare scenarios. In addition, it reveals methodological trends, such as the growing integration of machine learning and deep learning techniques, as well as persistent challenges related to data dependency, interpretability, scalability, and generalizability. These insights provide a conceptual foundation for the benchmarking framework introduced in Sect. 3.

2.4.1 Content-based treatment recommendations

Content-based treatment recommendation approaches aim to identify suitable therapeutic options by analyzing structured and unstructured patient-specific information, including medical history, clinical symptoms, diagnostic indicators, laboratory findings, and prior treatment responses. Unlike collaborative filtering techniques that rely on inter-patient similarities, content-based models operate independently for each patient, thereby reducing reliance on population-level behavioral data and mitigating data sparsity issues in scenarios with limited user interactions or newly diagnosed patients.

Despite these advantages, content-based approaches are highly dependent on the quality, completeness, and representativeness of feature engineering. They often struggle with cold-start conditions, limited adaptability to unseen clinical contexts, and reduced capacity to capture population-level treatment patterns [4]. Nevertheless, content-based models remain widely adopted in healthcare applications due to their transparency, interpretability, and direct alignment with patient-centric clinical attributes.

Content-based treatment recommendation methods can be broadly categorized into three principal approaches:

- *Feature similarity-based matching:* These approaches compute similarity scores between patient profiles and treatment characteristics using distance or similarity metrics such as cosine similarity, Euclidean distance, and TF-IDF weighting. They are particularly effective when structured patient records and well-defined treatment descriptors are available.
- *Machine learning-based classification:* Supervised learning models, including Support Vector Machines (SVM), Decision Trees, Multilayer Perceptrons (MLP), and deep neural networks, are employed to map patient conditions to optimal treatment options based on historical clinical outcomes and labeled medical data [5].
- *Association rule-based learning:* Pattern mining techniques such as Apriori and FP-Growth are used to identify frequent associations between patient conditions and effective treatments, thereby supporting evidence-driven clinical recommendations. (Appendix B, Algorithm 1). presents the pseudocode of a representative content-based filtering model used for treatment recommendation.

The effectiveness of content-based systems depends on robust modeling of both patient and treatment features. Patient modeling typically incorporates demographic attributes, comorbidities, genetic predispositions, physiological measurements, and medication adherence patterns. Treatment modeling emphasizes therapeutic efficacy, adverse effects, contraindications, and pharmacological properties. Figure 7 illustrates the workflow of a content-based treatment recommendation system, demonstrating how patient data are transformed into structured feature representations and matched with therapeutic options through relevance scoring mechanisms.

A comprehensive comparison of content-based treatment recommendation studies, including employed algorithms, disease domains, datasets, evaluation metrics, and reported limitations, is provided in Table 4. Analysis of these studies indicates that content-based approaches have achieved high reported accuracies in controlled experimental settings across a wide range of clinical applications, including oncology, mental health, cardiovascular disease, and chronic conditions. However, many existing studies

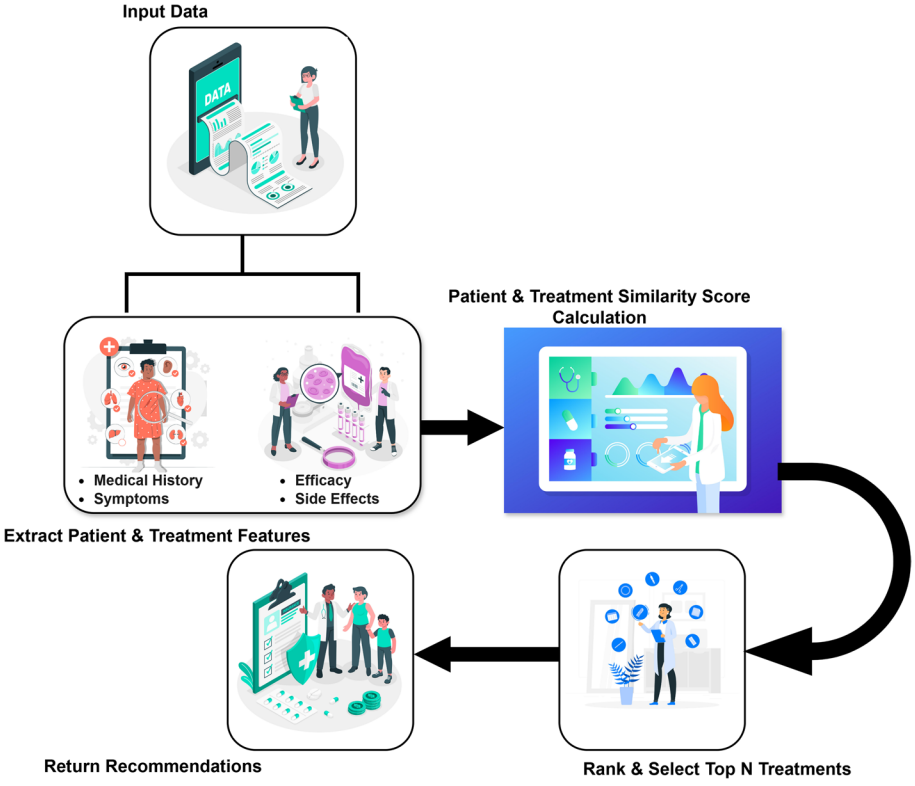


Fig. 7 Workflow of a content-based treatment recommendation system illustrating patient data extraction, feature representation, similarity computation, and relevance-based treatment selection

rely on retrospective datasets, exhibit limited external validation, and provide restricted interpretability, which can hinder clinical trust and real-world deployment.

Furthermore, the heterogeneity of datasets, feature representations, and evaluation protocols across studies complicates direct performance comparison and underscores the need for standardized benchmarking criteria. These limitations motivate the exploration of collaborative, knowledge-driven, and hybrid recommendation paradigms, which aim to address the inherent shortcomings of purely content-based approaches and are discussed in the subsequent subsections.

2.4.2 Collaborative filtering-based treatment recommendations

Collaborative filtering (CF) is a data-driven recommendation paradigm that leverages historical patient–treatment interaction patterns to infer suitable therapeutic options. Unlike content-based approaches, which rely solely on individual patient attributes, CF exploits collective intelligence by identifying similarities across patient populations or treatment responses, thereby enabling personalized recommendations based on shared clinical experiences.

Collaborative filtering techniques in healthcare are broadly categorized into patient-based and treatment-based approaches. Patient-based CF identifies clinically similar patients using medical histories, treatment outcomes, and demographic characteristics, and recommends therapies that have proven effective for comparable patient profiles. In contrast, treatment-based CF focuses on similarities among treatment options by analyzing response patterns, adverse effects, and efficacy trends observed across patient

Table 4 Systematic extraction for content-based (CB) personalized treatment recommendation studies

References	Year	Pub.	Source	Method / model	Clinical domain	Dataset characteristics	Metric(s)	Validation signal
[6]	2023	Conf.	IEEE	SVM + K-means	Stress alleviation (EEG)	Public EEG emotion dataset; 34 experiments; EMOTIV device	Acc. 96.3%	Benchmark-style evaluation on public dataset (external validation not stated)
[7]	2021	Conf.	IEEE	Signal processing + Random Forest (preference prediction)	Tinnitus (music therapy)	30 patients; 450 personalized music sessions	Acc. 73%	Small-cohort retrospective evaluation; confounders noted (mood/sleep)
[8]	2021	Conf.	IEEE	Rocchio + LambdaRank (IR-based ranking)	Cancer (clinical trial recommendation)	TREC 2017 clinical-trials track (benchmark)	nDCG = 0.58	Benchmark evaluation on shared task dataset (clinical validation not evaluated)
[9]	2024	Journal	IEEE	CNN + Beetle Antennae Search (BAS)	Liver cancer (nutritional support / rehabilitation)	1,000 patient samples (liver cancer cohort)	Acc. 92%	Retrospective single-cohort study; recurrence timing uncertainty may affect generalization
[10]	2022	Conf.	IEEE	NLP + NN + clustering (profiling + recommendation)	Dementia care support	∼200 patients (sample-size dataset stated)	Acc. 92%	Single-cohort evaluation; generalizability limitations stated
[11]	2023	Conf.	IEEE	Deep Attention Q-Network (Transformer-based RL)	ICU: sepsis / acute hypotension	MIMIC-III; 6,164 sepsis samples	Not stated	Retrospective EHR evaluation; dependence on longitudinal history noted
[12]	2024	Conf.	IEEE	CNN-based recommendation / planning	Colorectal tumor	TCGA cohort (size not stated)	Acc. 97%	Retrospective validation; authors note need for clinical trials
[13]	2020	Conf.	IEEE	DNN + SVM	Renal dialysis treatment planning	22,039 dialysis records; 180 patients	Acc. 95%	Historical-record evaluation; sensitivity to missing history noted
[14]	2022	Conf.	IEEE	Privacy-preserving PTR (HE + secure path protocol)	Chronic kidney disease	Not stated	Not stated	Security-focused evaluation; clinical validation details not stated
[15]	2023	Journal	Springer	Hybrid bio-inspired optimization (EHO + ALO) with multimodal inputs	Skin disease	HAM10000; 10,015 dermatoscopic images	Acc. 85.3%	Dataset-based evaluation; reliance on complete history highlighted

Table 4 (continued)

References	Year	Pub.	Source	Method / model	Clinical domain	Dataset characteristics	Metric(s)	Validation signal
[16]	2023	Conf.	IEEE	MIRNet + knowledge graphs (multi-visit)	Drug recommendation	MIMIC-III	Acc. 78.19%	Retrospective EHR validation; drug-interaction complexity caveat noted
[17]	2024	Conf.	IEEE	Explainable ML (gait-analysis PTR)	Parkinson's disease	Not specified	Acc. 98.01%	Dataset-based evaluation; interpretability/acceptance concerns noted
[18]	2020	Journal	IEEE	Q-Rank (algorithm ranking/integration)	Cancer (drug sensitivity)	NCI-DREAM7 benchmark	Acc. 89%	Benchmark comparison against baselines; transferability caveat stated
[19]	2023	Conf.	IEEE	Thompson Sampling (bandit-based PTR)	Not specified	Not specified	Not stated	Algorithmic study; interpretability/bias concerns stated; validation details not stated
[20]	2023	Conf.	IEEE	LR / DT / RF / XGBoost (multi-label pipeline)	Anxiety disorders	Clinical dataset; 1,369 patients	F1 up to 79.02%	Single-cohort clinical evaluation; evidence-alignment concern noted
[21]	2024	Journal	Springer	SCED infrastructure + experience sampling	Psychotherapy	Not specified	Not stated	Single-case design emphasis; empirical effectiveness evidence limited
[22]	2021	Journal	Springer	Regression vs. modified Kraemer method	Chronic fatigue syndrome	FINE trial; 296 patients	Acc. 95% (as reported)	Trial-derived evaluation; limited improvement vs TAU stated
[23]	2024	Journal	Springer Nature	Content-based filtering (diet adherence mediation)	Infertility treatment adherence	578 patients	Not stated	Retrospective/self-reported signals; recall bias noted
[24]	2024	Journal	Springer	PSM-based personalization analysis	Malnutrition / obesity nutrition intervention	20 households (rural Benin)	Not stated	Small N + attrition; propensity-score adjustment used
[25]	2023	Journal	MDPI	LIME-based explainable recommendation	Heart disease / diabetes	46,783 patients + diabetes dataset (768 records; Kaggle)	Acc. 98%	Dataset-based validation; interpretability highlighted
[26]	2023	Journal	Springer	RF / CatBoost / SVM	PPDM-A prediction	820 patients; 4 centers (China)	Acc. 92.7%	Multi-center retrospective validation; sample/selection bias noted

Table 4 (continued)

References	Year	Pub.	Source	Method / model	Clinical domain	Dataset characteristics	Metric(s)	Validation signal
[27]	2023	J	IEEE	Genetic algorithm (CB + genetic)	Skin disease	Cosing DB (EU Open Data Portal)	Acc. 71%	Genotype-centric; ignores environment/allergies; limited phenotype integration
[28]	2024	Journal	PubMed	Decision-making algorithm (EHR discharge management)	Diabetes	Single-center; 835 patients	Not stated	Single-center; no control group (generalization limited)
[29]	2022	Journal	PubMed	Weighted kNN (statin intensity recommendation)	ASCVD / statin therapy	EHR; 50,576 patients	Acc. 83.83%	Cohort heterogeneity; high-risk restriction not enforced
[30]	2023	Journal	Sci. Direct	LR+RFE (decision aid)	Prostate cancer	PCPR registry	Acc. 76%	Registry-based evaluation; clinical adequacy caveat stated
[31]	2024	Journal	MDPI	Q-learning (policy)	Hypertension / T2D	NHISS (Korea)	Acc. 78.7%	Population dataset; cross-setting generalizability not assessed
[32]	2023	Journal	Wiley	RF+LR (CBT benefit prediction)	Psychological disorders	$n>1,300$ (naturalistic cohort)	Acc. 91%	Holdout caveat; not significant across the full sample
[33]	2023	Journal	Springer	LR / NN / RF / CatBoost / SVM	Pancreatitis diabetes	820 patients; 4 centers	Acc. 92.7%	Retrospective multi-center; bias/generalization caveat
[34]	2023	Journal	Springer	Balanced Random Forest	Breast cancer psychosocial outcomes	BOUNCE consortium	Acc. 85%	Consortium dataset; single-consortium generalization caveat
[35]	2022	Journal	Springer	DL segmentation + radiomics	Low-grade glioma survival prediction	MRI + clinical data; 349 patients	C-index 0.82; (Acc. 95% in notes)	Retrospective cohort; imaging heterogeneity caveat
[36]	2024	Journal	IEEE	DRecHGR (heterogeneous graph model)	Multimorbidity (drug recommendation)	Not specified	Acc. 91.82%	EHR network evaluation; confounding/generalizability caveat stated
[37]	2023	Journal	Springer	LLM-based recommendation (GPT-4)	Orthopedic knee/shoulder conditions	20 anonymized MRI reports	Not stated	Expert-rated outputs; urgency/context limitation noted

Table 4 (continued)

References	Year	Pub.	Source	Method / model	Clinical domain	Dataset characteristics	Metric(s)	Validation signal
[38]	2023	Journal	Springer	Lasso / Ridge / RF / LightGBM (personalized antibiograms)	UTIs / antibiotic selection	Stanford EHR 8,342; Boston EHR 15,806	Coverage 90.4%	Two-site evaluation; modest discrimination reported
[39]	2023	Journal	Springer	Multi-criteria decision framework (logical model)	Preventive / personalized medicine	Not stated	Not stated	Qualitative framework; subjectivity/consistency caveat noted
[40]	2021	Conf	IEEE	PCPRec: LSTM/BiLSTM/GRU + self-attention	Coronary heart disease	MIMIC-III (995 patients)	Acc. 91.02%	Patient representation limited (physical signs); broader features not integrated

Reported metrics and validation descriptors are taken from the study-reported evaluation as summarized in this survey ("Not stated" indicates missing information in the source description)

cohorts. (Appendix B, Algorithm 2) presents the pseudocode of a representative collaborative filtering–based treatment recommendation framework [41].

One of the key strengths of CF lies in its ability to adapt dynamically to real-world clinical environments by learning from large-scale longitudinal patient data. By incorporating information such as treatment adherence, symptom progression, follow-up outcomes, and patient-reported outcome measures (PROMs), CF models can capture gradual changes in treatment effectiveness over time. This temporal learning capability supports continuous refinement of personalized treatment strategies.

From a methodological perspective, CF models rely on high-dimensional similarity computations using measures such as cosine similarity and Pearson's correlation coefficient. Patient-based CF derives recommendations from historical outcomes of clinically similar patients, whereas treatment-based CF clusters therapies with overlapping effectiveness profiles to generate evidence-informed treatment rankings [42]. However, these approaches are inherently sensitive to data sparsity, missing values, and cold-start scenarios, particularly in early-stage diagnosis or rare disease settings.

To mitigate sparsity and scalability challenges, recent studies increasingly integrate matrix factorization techniques such as Singular Value Decomposition (SVD) and Non-Negative Matrix Factorization (NMF), as well as deep learning architectures including autoencoders and graph neural networks. Reinforcement learning enhanced CF models further enable closed-loop optimization by continuously updating treatment recommendations based on observed patient outcomes. Figure 8 illustrates the workflow of a collaborative filtering–based treatment recommendation system, highlighting similarity computation and relevance-based treatment ranking using historical patient interactions (Table 5).

A detailed comparative summary of collaborative filtering–based treatment recommendation studies, including algorithmic variations, disease domains, datasets, evaluation metrics, and reported limitations, is provided in Table 7. Analysis of these studies indicates that CF approaches achieve strong predictive performance in data-rich clinical domains such as cardiovascular disease, diabetes, and mental health. Nevertheless, their

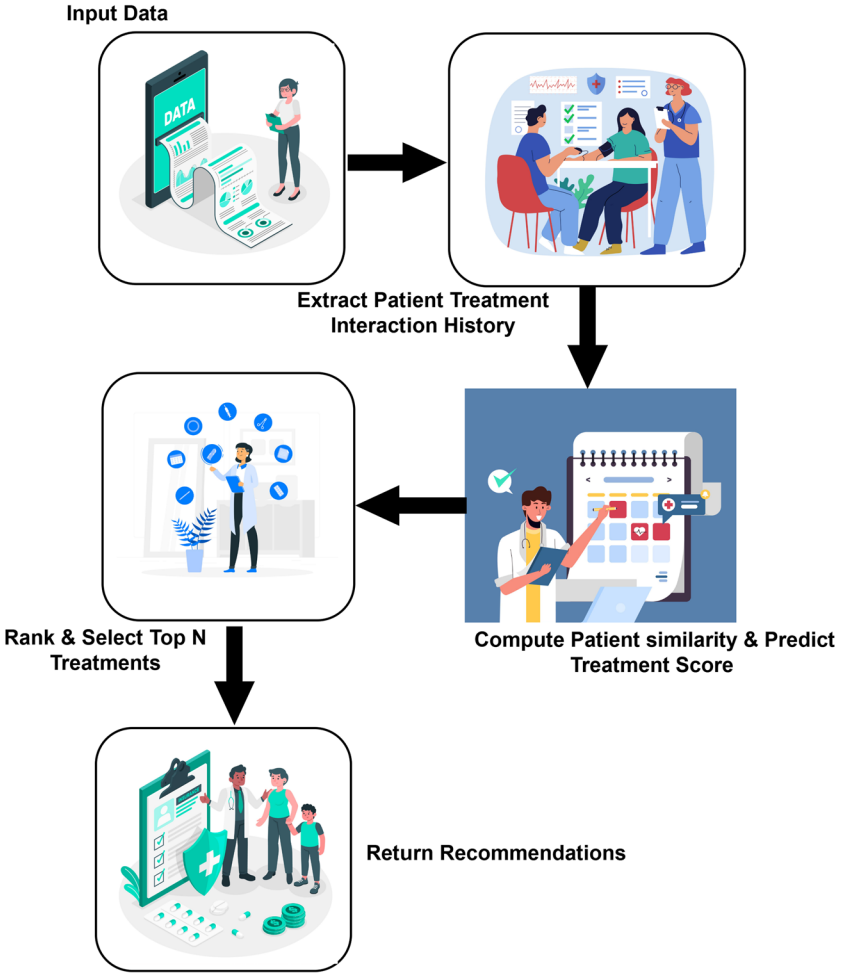


Fig. 8 Systematic extraction of collaborative-filtering (CF) treatment recommendation studies summarizing clinical domain, dataset characteristics, validation design, and reported performance indicators

broader clinical adoption is constrained by dependence on large, high-quality patient histories, limited interpretability of complex models, and privacy concerns associated with sensitive medical data.

These limitations motivate the incorporation of domain knowledge and hybrid recommendation paradigms that seek to balance personalization accuracy with robustness, interpretability, and clinical trustworthiness, as discussed in the subsequent subsections.

2.4.3 Knowledge-based filtering treatment recommendations

Knowledge-Based Filtering (KBF) represents a clinically grounded recommendation paradigm that integrates expert knowledge, medical guidelines, and patient-specific information to support evidence-based treatment decisions. In contrast to content-based and collaborative filtering approaches that primarily rely on historical patient interactions, KBF directly encodes domain knowledge derived from clinical protocols, expert rules, and validated medical evidence, thereby ensuring alignment with established standards of care [51].

KBF systems typically operate through two complementary perspectives: patient-centric and treatment-centric reasoning. Patient-centric KBF analyzes individual patient

Table 5 Systematic extraction table for reviewed studies: collaborative filtering (CF) paradigms

References	Year	Type	Source	Method / Model	Clinical domain	Dataset / cohort	Metric(s)	Validation signal
[43]	2023	Conf.	IEEE	Q-learning (RL)	Diabetes	South Korea EHRs	NR	Ethics and governance implications of patient-data use not fully addressed
[44]	2024	Conf.	IEEE	DL-D + RL-MRO	Mental health (multi)	fMRI images	Acc. 91%	Implementation and real-world deployment considerations not fully evaluated
[45]	2023	Journal	ProQuest	RE-loc-eb (neighborhood CF)	Hypertension	EHR (298 patients)	Acc. 96%	Outcome ground truth limited; agreement-based evaluation may not capture causal benefit
[46]	2022	Journal	Elsevier	SAI-DQN (knowledge-guided DQN)	Sepsis (antibiotic combos)	MIMIC-IV (retrospective)	Survival 77.0%	Retrospective database; missing clinical indicators may limit optimal regimen inference
[47]	2024	Journal	Elsevier	MLP + RSF (counterfactual + MO)	ASCVD (statin planning)	Optum Labs DW (>126.6M)	NR	Multi-arm balance remains challenging; residual confounding may affect robustness
[48]	2019	Journal	Elsevier	EMA-based modeling	Cancer-related fatigue	Five adult cases (CCRF)	NR	Small N; feedback interpretation and clinical integration challenges reported
[49]	2022	Journal	Scopus	SmartPLS (modeling)	Cancer (general)	NR	NR	High-level claims; limited empirical benchmarking and deployment evidence
[50]	2022	Journal	MDPI	RBM-CNN (CF)	Infectious diseases	UCI healthcare dataset	RMSE 2.70	K-sensitivity trade-off; privacy/utility risks when neighborhood size increases

profiles including medical history, comorbidities, risk factors, and therapeutic goals and maps these attributes to validated clinical guidelines to generate personalized treatment recommendations. Treatment-centric KBF, on the other hand, evaluates therapeutic options by leveraging evidence from clinical trials, expert-curated knowledge bases, and historical treatment outcomes to identify interventions with demonstrated efficacy for specific disease populations. (Appendix B, Algorithm 3) outlines a representative knowledge-based filtering framework for individualized treatment recommendation.

A key strength of KBF lies in its strict adherence to evidence-based medical practice, which enhances clinical reliability and interpretability. By explicitly modeling clinical rules, contraindications, and safety constraints, KBF systems can provide transparent recommendations that are easier for clinicians to validate and trust [52]. However, their performance is strongly influenced by the quality, coverage, and interoperability of the underlying knowledge base. Effective deployment therefore requires seamless integration with electronic health records (EHRs), clinical decision support systems, and continuously updated medical knowledge repositories.

To improve adaptability and scalability, recent KBF frameworks increasingly incorporate machine learning–driven refinement mechanisms, natural language processing (NLP), and ontological reasoning. These enhancements enable systems to assimilate emerging medical literature, update expert rules dynamically, and reason over complex clinical relationships. Such hybridized KBF models are particularly valuable in domains such as precision oncology, rare disease management, and personalized nutrition therapy, where guideline adherence and expert oversight are critical.

Figure 9 illustrates the workflow of a knowledge-based treatment recommendation system, demonstrating how patient data, expert knowledge, and clinical evidence are synthesized to generate optimized therapeutic suggestions.

A detailed comparison of knowledge-based treatment recommendation studies including applied algorithms, disease domains, datasets, performance metrics, and reported limitations is provided in Table 6. Existing studies demonstrate that KBF approaches achieve reliable performance in guideline-intensive clinical settings, such as diabetes management, chronic disease care, and decision support for medical nutrition therapy. Nevertheless, their broader adoption remains constrained by limited coverage of complex multimorbidity scenarios, high knowledge engineering costs, and the need for continuous expert validation.

Furthermore, research on KBF-based personalized treatment systems remains relatively limited to specialized domains such as rare diseases, mental health, pediatric and geriatric care, and integrative medicine. These limitations stem from regulatory and ethical constraints, scarcity of high-quality annotated clinical data, and challenges in integrating heterogeneous medical knowledge sources. Addressing these challenges motivates the development of hybrid recommendation paradigms that combine knowledge-driven reasoning with data-driven learning, as discussed in the following subsection.

2.4.4 Hybrid treatment recommendations

Hybrid Treatment Recommendation (HTR) systems represent an advanced paradigm in personalized medicine by integrating multiple recommendation strategies to overcome the inherent limitations of standalone approaches. In contrast to isolated content-based,

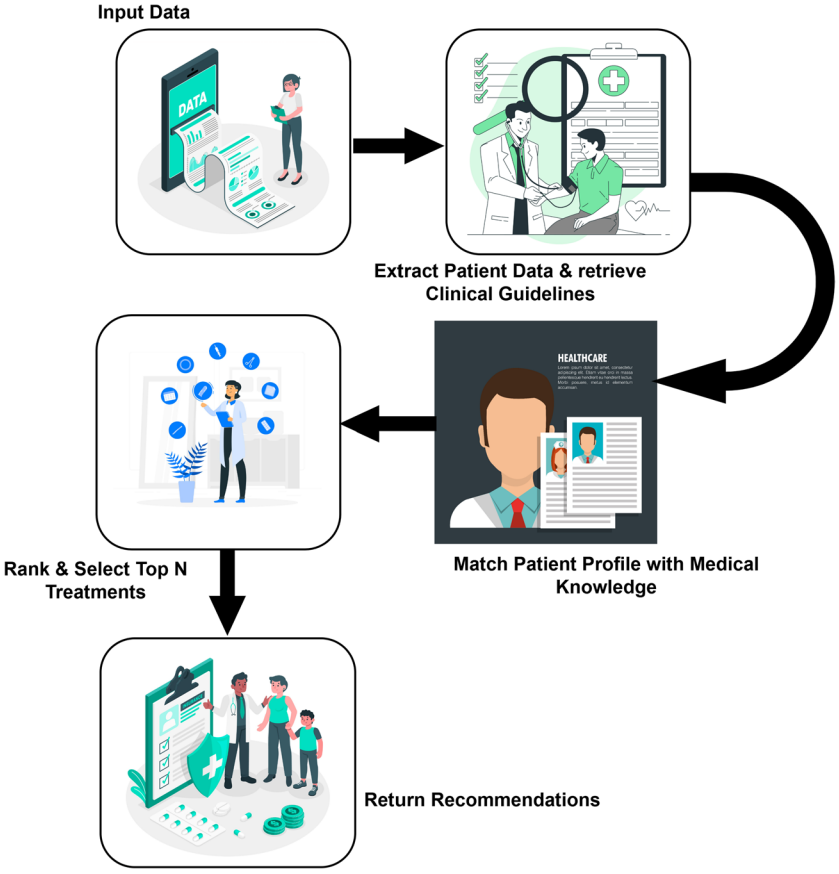


Fig. 9 Clinical knowledge–based filtering workflow illustrating the integration of patient data, expert knowledge, and medical guidelines to support personalized treatment recommendations

collaborative filtering, or knowledge-based models which often suffer from data sparsity, cold-start issues, limited adaptability, or restricted clinical coverage hybrid systems synergistically combine these techniques to enhance recommendation robustness, precision, and clinical relevance [56].

The primary objective of HTR in healthcare is to improve patient outcomes, increase treatment adherence, and support efficient clinical decision-making. Unlike consumer-oriented recommender systems where preferences change rapidly, patient behavior in healthcare is longitudinal, shaped by chronic disease trajectories, lifestyle modifications, and evolving therapeutic responses. Hybrid models therefore integrate patient-specific data with evidence-based treatment guidelines, medication protocols, behavioral interventions, and contextual clinical knowledge to maintain sustained relevance across the care continuum.

Hybrid treatment recommendation systems are commonly realized through three architectural strategies [57–59]:

- *Monolithic hybrid models:* These systems unify multiple recommendation mechanisms within a single framework by jointly modeling patient attributes, clinical features, and heterogeneous data sources. By performing deep feature fusion, monolithic hybrids enhance predictive accuracy but may incur higher computational and interpretability costs.

Table 6 Systematic extraction of knowledge-based (KB) treatment recommendation studies highlighting rule/knowledge sources, clinical context, evaluation setting, and reported performance or agreement indicators

References	Year	Type	Source	Method / Model	Clinical domain	Dataset / cohort	Metric(s)	Validation signal
[53]	2022	Journal	Springer	Similarity-metric KB filtering	Chronic disease (general)	User study (60 participants)	Acc. 70%	Narrow focus on health behaviors; limited recommendation granularity without human coaching; calls for richer user models and stronger empirical validation
[54]	2022	Journal	IEEE	Rule-based + user profiles	Diabetes (community-specific)	NR	Acc. 92.4%	Evaluation relies on use-cases/expert verification; lacks full-scale testing with target population
[55]	2024	Journal	Sci. Direct	OnT2D CDSS *(RF/SVM/KNN/NB)*	Type 2 diabetes (MNT)	Medical Nutrition Therapy (MNT) data	Acc. 87%	Requires broader validation and clinical expert fine-tuning for real-world effectiveness

- *Parallel hybrid models:* Multiple recommendation models are executed simultaneously, and their outputs are combined using weighted aggregation, ensemble learning, or dynamic switching strategies. This architecture is particularly effective in adapting to real-time patient data and contextual changes, enabling responsive and personalized treatment planning.
- *Pipelined hybrid models:* These systems apply recommendation techniques sequentially, where each stage refines the candidate treatment set using cascading filters or meta-learning. Pipelined hybrids improve efficiency by progressively narrowing the search space while preserving clinically relevant options. (Appendix B, Algorithm 4) presents the pseudocode of a representative hybrid filtering–based treatment recommendation framework.

By integrating complementary mechanisms (e.g., patient-feature similarity, population-level patterns, and guideline/constraint checks), hybrid PTR systems are typically designed to improve robustness under heterogeneous data conditions and to better align recommendations with clinical constraints. In the reviewed corpus, many hybrid studies report improvements over at least one stated single-paradigm baseline within their own experimental setup; however, the strength and generality of these gains vary with the fusion design, baseline strength, and validation depth, and therefore do not support a universal claim of superiority across all contexts. Figure 10 presents a representative hybrid PTR workflow that combines content-based, collaborative, and knowledge-based components within a single decision-support pipeline, highlighting how multiple evidence sources can be reconciled prior to producing a treatment suggestion.

A detailed comparative overview of hybrid treatment recommendation studies including employed hybridization strategies, disease domains, datasets, evaluation metrics, and reported limitations is provided in Table 7.The reviewed hybrid studies frequently report improvements over at least one stated single-paradigm baseline within their own experimental setup; however, the magnitude and generality of these gains vary with fusion design, baseline strength, and validation depth. Accordingly, the corpus does not support a universal claim that hybrid systems outperform single-paradigm approaches across tasks and settings.

Despite these advantages, hybrid models face practical challenges related to system complexity, data integration across heterogeneous sources, and limited generalizability beyond controlled experimental settings. Recent studies also highlight emerging risks, such as recommendation hallucinations in conversational or generative hybrid systems, which may undermine clinical trust if not adequately addressed. Overcoming these challenges requires standardized evaluation practices, interpretability-aware model design, and rigorous clinical validation to ensure safe, reliable, and trustworthy deployment in real-world healthcare environments.

3 Summary of key findings from personalized treatment recommendation systems

This section synthesizes the key findings derived from the systematic analysis of personalized treatment recommendation systems (PTRS), with an emphasis on methodological trends, therapeutic focus, disease coverage, and algorithmic adoption. Rather than presenting isolated performance results, the analysis consolidates insights across

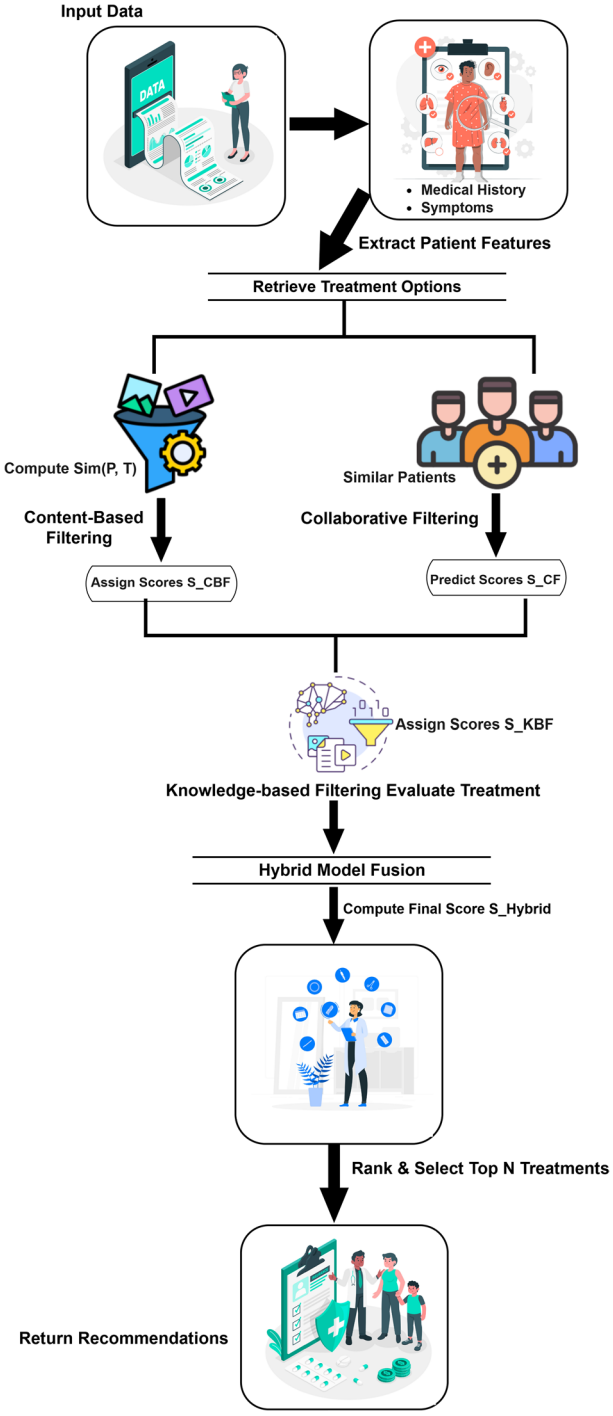


Fig. 10 Hybrid treatment recommendation workflow integrating content-based, collaborative, and clinical knowledge–based filtering to generate unified relevance scores for personalized treatment selection

heterogeneous studies to identify dominant patterns, recurring limitations, and emerging research directions.

The reviewed literature demonstrates that PTRS approaches including content-based, collaborative filtering, knowledge-based, and hybrid models exhibit complementary strengths and limitations depending on data availability, clinical context, and treatment objectives. Content-based methods constitute the largest share of the included corpus

Table 7 Systematic extraction of hybrid treatment recommendation studies summarizing fused paradigms, data modalities, validation approach, and reported performance indicators across clinical domains

References	Year	Type	Src.	Method / Model	Clinical domain	Dataset / cohort	Metric(s)	Validation signal
[60]	2024	C	IEEE	SVM + genetic (CB+CF)	Metabolic syndromes	USDA National Nutrient Database	NR	Deployment and user engagement not validated; performance not reported
[61]	2022	J	Springer	DT + genetic (KB + data-driven CDSS)	Drug-resistant TB	Exome seq. 355 RR-TB (129 profiles)	P@1 = 95%	Overfitting noted; weaker performance under external validation
[62]	2023	J	Scopus	Hybrid recommender (multi-technique)	Mental health	Demographic data (180 patients)	Acc. 83%	Integration complexity (data + algorithm selection); implementation burden
[63]	2023	J	MDPI	SHARE: content + collaborative	Diet / chronic diseases	Food.com recipes (2,774,676)	Acc. 89%	Depends on user engagement; real-time input variability may affect quality
[64]	2023	J	PubMed	KNN/SVM/RF (hybrid CB+CF)	Type 2 diabetes	UK Biobank (>500k)	Acc. 99.4%	Needs RCT/real-world validation to confirm sustained clinical impact
[65]	2019	J	PubMed	DPAC (CAMIL), content-based + Bayesian network	Breast cancer	LSDS-5YDM dataset (single-site study)	Acc. 89%	Single-center dataset; no external validation reported.
[66]	2023	J	Sci. Dir.	KM2PS interaction-aware knowledge model	Pharmacogenomics	DDI, DGI, and DHI knowledge resources	NR	Validated on limited test cases; generalizability remains unclear.
[67]	2023	J	Sci. Dir.	Hybrid RF, SVM, KNN, NB, XGB, and ANN models	Cardiovascular risk	ARIC cohort with class imbalance	Acc. 70%	Class imbalance may affect reliability; mitigation strategies not detailed.
[68]	2024	J	Sci. Dir.	ChatDiet recommender using BM25 retrieval	Cardiovascular and nutrition	N-of-1 longitudinal study (3-year, single subject)	Eff. 95%	Hallucinations observed; single-subject design limits external validity.
[69]	2024	J	Sci. Dir.	Selector-based drug ranking using SVM and NN	Cancer	REVERT dataset (845 oncology patients)	SVM 64.1%, NN 65.4%	Retrospective evaluation only; population-level transferability unclear.
[70]	2019	J	MDPI	SRHL hybrid learning recommender	Obesity, diabetes, and CVD	USDA Branded Food Products Database	Err. 3.65, Acc. 94.16%	Reported gains modest at organizational scale; scalability limitations noted.

(Fig. 11). This pattern may reflect practical factors such as wider availability of structured patient covariates than dense interaction logs, methodological accessibility, and publication practices, rather than an inherent superiority across tasks. Pharmacological treatments dominate current research, underscoring the importance of individualized medication planning, while non-pharmacological interventions such as lifestyle modification and behavioral therapy remain comparatively underexplored despite their clinical relevance.

Across disease categories, chronic conditions particularly diabetes and cardiovascular diseases receive the greatest research attention, reflecting the long-term management requirements and the need for adaptive, patient-centric decision support. From a methodological perspective, machine learning and deep learning techniques increasingly underpin PTRS, enhancing predictive accuracy and adaptability. However, the diversity of datasets, clinical settings, and evaluation protocols across studies necessitates careful interpretation of comparative trends, motivating a qualitative and multi-dimensional synthesis rather than direct numerical comparison.

3.1 Overview of techniques in personalized treatment recommendations

To reduce ambiguity, terms are used consistently throughout the survey. Paradigm denotes the high-level recommendation family: content-based (CB), collaborative filtering (CF), knowledge-based (KB), or hybrid. Model/algorithm refers to the specific computational method used within a paradigm (e.g., RF, SVM, LSTM, DQN). Technique denotes a component-level mechanism (e.g., attention, metric learning, feature attribution). Design strategy refers to an architectural choice (e.g., early/late fusion, rule-constrained recommendation). Descriptive statements report corpus properties, whereas interpretive statements are explicitly qualified (e.g., "may", "is consistent with") unless directly supported by study-level evidence.

Recent advances in personalized treatment recommendation (PTR) research can be broadly categorized into four primary methodological paradigms: content-based (CB), collaborative filtering (CF), knowledge-based (KB), and hybrid recommendation approaches. Each paradigm reflects a distinct strategy for representing patient state and treatment decision logic, resulting in different strengths and constraints across clinical settings.

Content-based techniques aim to recommend therapeutic options by aligning treatment attributes with individual patient characteristics such as medical history, physiological indicators, and preferences. Their ability to operate independently of population-level interaction data allows them to mitigate data sparsity and cold-start challenges, making them particularly suitable in clinical environments where patient–treatment interaction records are limited. However, their reliance on well-structured feature representations can restrict generalization and limit the capture of broader population-level treatment patterns.

Collaborative filtering–based methods generate recommendations by identifying similarities among patients or treatments based on historical treatment responses. By leveraging collective clinical experience, CF approaches can uncover latent treatment relationships that are not evident from individual profiles alone. Despite this advantage, CF systems remain vulnerable to cold-start issues, incomplete patient histories, and sparsity in medical interaction data, especially in rare disease or early-diagnosis settings.

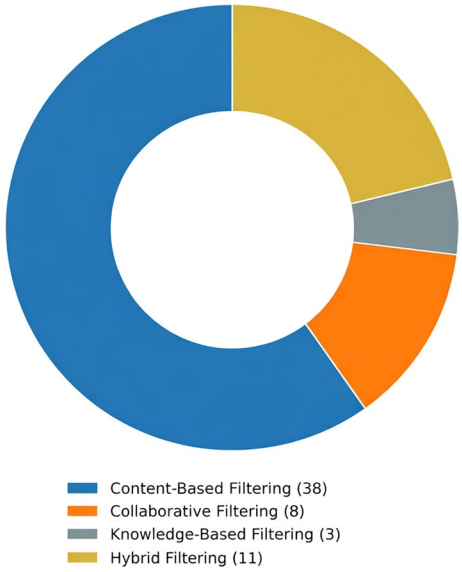


Fig. 11 Distribution of reviewed studies across recommendation techniques, showing the relative prevalence of content-based, collaborative, knowledge-based, and hybrid filtering approaches

Hybrid recommendation methods integrate content-based and collaborative strategies to balance personalization and robustness. By combining multiple information sources and decision mechanisms, hybrid systems are commonly proposed to mitigate limitations of single-paradigm designs and may improve robustness in complex or heterogeneous care settings, depending on fusion design and validation depth. In contrast, knowledge-based systems rely on expert-defined rules, clinical guidelines, and medical ontologies to recommend treatments consistent with established standards of care, thereby supporting interpretability and clinician-facing rationale.

The distribution of reviewed studies across recommendation paradigms is summarized in Fig. 11. Content-based approaches constitute the largest share of the surveyed corpus. This prevalence is reported here as a descriptive feature of the included literature and may be influenced by pragmatic factors such as the wider availability of structured patient covariates relative to dense treatment–outcome interaction logs, methodological accessibility, and venue-specific publication practices. In many clinical settings where comprehensive longitudinal interaction histories are limited, content-based designs are feasible because they operate directly on patient-level features without requiring population-scale feedback matrices.

Hybrid and knowledge-based systems appear less frequently, but they reflect an ongoing effort to combine complementary information sources (e.g., clinical guidelines, safety constraints, ontologies, and multi-modal inputs) to address sparsity and to align recommendations with workflow and safety requirements. Importantly, reported gains for these systems are typically demonstrated within study-specific tasks and validation setups; therefore, cross-paradigm comparisons should be interpreted cautiously because endpoints, datasets, and evaluation protocols remain heterogeneous across the corpus.

3.2 Types of therapies and related studies

In contemporary healthcare, patient-specific data–driven systems are increasingly employed to support personalized treatment planning across a broad spectrum of

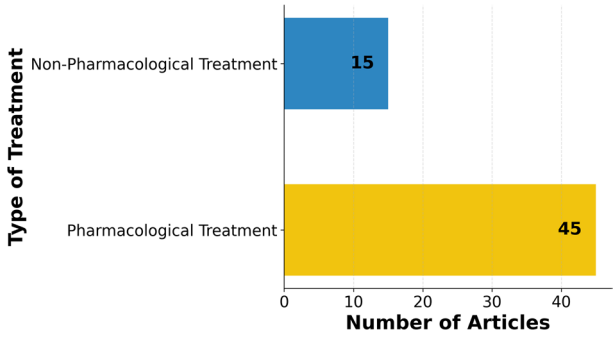


Fig. 12 Distribution of reviewed studies by treatment type, comparing pharmacological and non-pharmacological treatment approaches based on the number of published articles

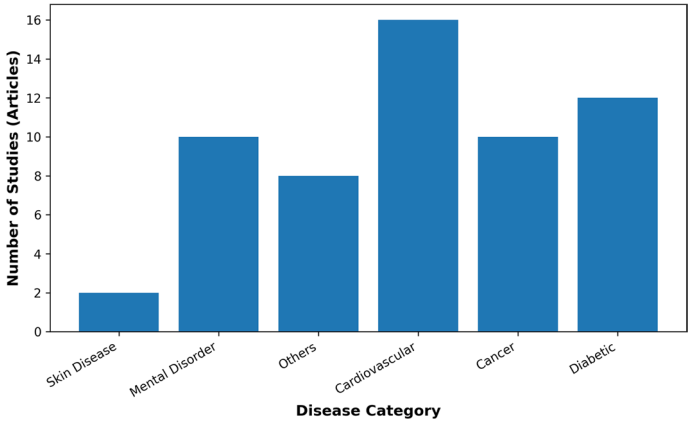


Fig. 13 Distribution of disease categories in the reviewed studies, illustrating overall disease coverage and a focused sub-classification of chronic conditions

therapeutic categories, encompassing both pharmacological and non-pharmacological interventions. Personalized treatment recommendation systems (PTRS) aim to assist clinicians in selecting optimal therapies tailored to individual patient profiles, thereby enhancing clinical decision-making and treatment effectiveness.

Pharmacological treatments remain the predominant focus in the reviewed literature, as personalized recommendation frameworks increasingly leverage patient-specific attributes such as genetic information, clinical history, and prior medication responses to optimize drug selection and dosage while minimizing adverse effects. In parallel, non-pharmacological interventions including physical therapy, psychological counseling, behavioral modification, and lifestyle interventions are recognized as essential components of holistic patient care. PTRS assess patient preferences, behavioral patterns, and longitudinal health data to recommend non-drug strategies that improve adherence, patient engagement, and long-term health outcomes.

As illustrated in Fig. 12, a substantial proportion of the reviewed studies concentrate on pharmacological treatments, underscoring the central role of medication personalization in current research. This emphasis reflects both the availability of structured pharmacological data and the clinical importance of individualized drug regimens. Although non-pharmacological interventions are less frequently explored, they offer valuable complementary strategies that support sustainable health improvements and patient-centered care.

Disease-wise analysis further reveals a strong emphasis on chronic conditions within personalized treatment recommendation research. As shown in Fig. 13, diabetes and cardiovascular diseases dominate the literature, reflecting their high prevalence, long-term management requirements, and substantial clinical burden. Personalized treatment strategies in these domains often combine pharmacological optimization with lifestyle and behavioral interventions. In contrast, acute conditions and less prevalent diseases such as infections, dermatological disorders, and influenza receive comparatively limited attention, likely due to smaller patient cohorts and shorter treatment horizons. Increased research attention is also observed in mental health conditions, including anxiety and hypertension, highlighting the growing recognition of personalized interventions in these areas.

From a methodological perspective, Fig. 14 illustrates the temporal evolution of artificial intelligence techniques applied in personalized healthcare between 2019 and 2024. Machine learning approaches dominate the landscape, with Random Forest models appear frequently in the surveyed studies. This usage may be consistent with their practicality for structured clinical features and their relatively straightforward training and inspection compared with more complex architectures. Support Vector Machines (SVM) and k-Nearest Neighbors (KNN) are also frequently employed, reflecting their effectiveness in diagnostic and predictive tasks. More recently, deep learning models such as multilayer perceptrons and convolutional neural networks have gained prominence, particularly for complex data modalities including medical imaging. Notably, reinforcement learning techniques have seen increased adoption since 2021, with Deep Q-Networks (DQN) enabling adaptive, outcome-driven treatment recommendation strategies.

3.3 Major insights and outcomes of personalized treatment recommendation systems

Personalized treatment recommendation systems (PTRS) are increasingly explored as decision-support tools that integrate heterogeneous patient information (e.g., clinical records, genomic signals, and stated preferences) to generate patient-specific treatment suggestions. Within the included corpus, content-based and hybrid paradigms constitute the largest shares (Fig. 11). This distribution is a descriptive property of the reviewed literature and should not be interpreted as evidence of paradigm superiority; it may reflect practical factors such as the relative availability of structured patient

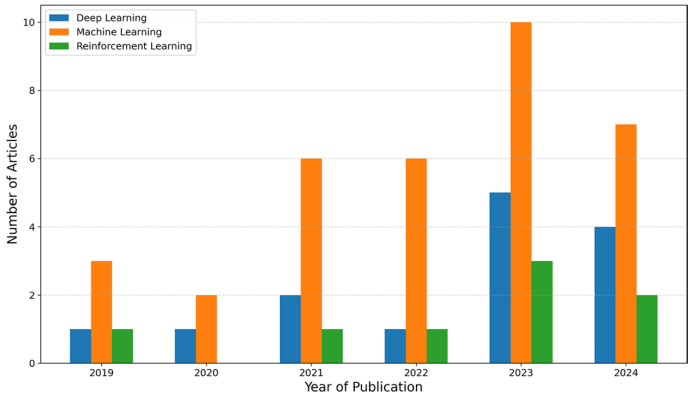


Fig. 14 Year-wise distribution of reviewed studies across learning paradigms, illustrating trends in deep learning, machine learning, and reinforcement learning approaches from 2019 to 2024

covariates compared with dense treatment–outcome interaction logs, common retrospective validation setups, and differences in implementation and reporting complexity across paradigms.

Content-based systems operationalize personalization by matching treatment options to explicit patient attributes and clinically relevant features. This design is frequently adopted in settings where interaction matrices are sparse or unavailable because it can operate directly on patient-level covariates without requiring population-scale feedback signals. Hybrid systems combine multiple paradigms (e.g., feature-based similarity with interaction-driven inference and/or guideline constraints) and are commonly motivated by clinical tasks that involve heterogeneous inputs, multimorbidity, or temporal disease trajectories. Reported benefits of hybridization are typically study-specific (often relative to at least one baseline within the same experimental setup) and depend on fusion design choices and validation depth rather than reflecting a corpus-level guarantee of improved performance.

Across paradigms, the increasing use of advanced AI components—including deep learning for representation learning and reinforcement learning for sequential decision support is associated with greater modeling flexibility in data-rich scenarios (e.g., longitudinal EHR settings). At the same time, higher model complexity introduces recurring barriers to translational readiness, particularly when studies provide limited detail on model specification, robustness checks, or generalization testing. These limitations complicate cross-study comparability and can restrict clinical uptake when decision logic is difficult to audit.

A further recurring issue is the reliance on opaque or weakly explained models, which can impede clinician trust and hinder deployment in safety-critical workflows. In addition, many PTRS depend on high-quality, consistently captured patient data, which may vary across institutions and patient populations. Strengthening interpretability, improving reporting transparency, and adopting more rigorous validation designs (e.g., temporally separated testing, external cohorts, or multi-site evaluation when feasible) remain central requirements for translating PTRS from research prototypes into reliable, clinically usable systems.

3.4 Benchmarking and evaluation strategies in PTR systems

Robust benchmarking is central to establishing whether AI-based personalized treatment recommendation (PTR) systems are not only accurate, but also clinically actionable and deployable. Our review nevertheless reveals substantial variation in evaluation practice. Studies draw on heterogeneous data sources (e.g., proprietary EHRs, disease registries, public datasets, and synthetic/simulation data) and frequently report conventional predictive metrics (e.g., accuracy, AUC, F1-score) without sufficient clinical contextualization. Consequently, cross-study comparison is often confounded by differences in cohort definition, outcome specification, treatment/action space, and validation protocol, while clinically critical facets such as safety, actionability, interpretability, and deployment constraints are inconsistently documented.

To enable principled synthesis under this heterogeneity, we complement metric-centric reporting with a structured rubric-based assessment (Appendix A; Table 9) that summarizes *evidence strength* along methodological and translational dimensions. This rubric is not intended to rank studies by absolute performance values, which are

rarely comparable across datasets, endpoints, and treatment spaces, but to standardize how evaluation rigor and clinical readiness are reported and interpreted across the surveyed corpus. Operational definitions and anchor criteria are specified in (Appendix A, together with the scoring workflow (independent scoring with justification notes followed by structured adjudication) and worked, evidence-linked examples (Appendix A; Table 10), to make the assessment auditable.

The rubric comprises one descriptive taxonomy field and five scored dimensions:

- *Recommendation Paradigm (descriptive):* Records the underlying personalization strategy (content-based, collaborative filtering, knowledge-based, or hybrid). This field supports stratified synthesis and is not included in the aggregate score.
- *Algorithmic Rigor (scored):* Captures methodological transparency and robustness, including clarity of model specification, justified design choices, baseline selection, and robustness checks (e.g., ablation/sensitivity analysis).
- *Clinical Applicability (scored):* Assesses clinical actionability and workflow alignment, including whether outputs map to explicit clinical decision points and whether deployment-relevant constraints (e.g., usability, safety monitoring) are considered.
- *Validation Depth (scored):* Reflects the strength of generalization evidence, distinguishing weak or poorly specified evaluation from retrospective validation with appropriate partitioning, external validation on independent cohorts, multi-site evaluation, or prospective assessment.
- *Data Transparency and Ethics (scored):* Evaluates the completeness of data provenance and preprocessing reporting and explicit attention to governance/privacy and ethics-related safeguards where applicable.
- *Scalability and Explainability (scored):* Assesses operational feasibility and interpretability in clinically meaningful terms, including computational constraints, generalization limits, integration feasibility, and clinician-interpretable explanations.

Each scored dimension uses a 10-point ordinal scale, and an aggregate score (mean across the five scored dimensions) is reported for descriptive synthesis. Where strata are reported, they are framed as *evidence-strength groupings* to support trend analysis rather than definitive quality ranking; thresholds are specified in (Appendix A; Table 9 footnote). Importantly, rubric-derived strata are not treated as stand-alone evidence for claims of paradigm superiority: comparative statements about paradigm effectiveness are restricted to studies that provide direct head-to-head evaluations under comparable validation conditions.

Overall, this benchmarking strategy prioritizes auditability, clinical grounding, and methodological consistency, enabling corpus-level synthesis while explicitly acknowledging the constraints imposed by heterogeneous data sources, modeling assumptions, and evaluation protocols.

3.5 Determinants of model effectiveness across clinical contexts

Cross-domain synthesis indicates that paradigm effectiveness is shaped by the alignment between (i) the clinical decision setting, (ii) data availability and structure, and (iii) deployment constraints. However, because the reviewed studies optimize heterogeneous endpoints (e.g., Acc/AUC/F1, survival/outcome rates, ranking metrics such as nDCG/RMSE, agreement/concordance, or task-specific utility measures) and apply different

validation protocols, raw performance values are rarely comparable across papers. To make the evidentiary basis of our synthesis auditable, Appendix A reports systematic extraction Tables 4, 5, 6 and 7, recording (as stated in each primary study) the paradigm, clinical domain, dataset descriptors, validation descriptors, and the primary evaluation indicator(s). Rubric-derived strata are used for descriptive synthesis of reporting/validation patterns and are not treated as stand-alone evidence of paradigm superiority; comparative effectiveness statements are restricted to studies reporting direct head-to-head evaluations under comparable validation conditions.

3.5.1 Quantitative synthesis from the extracted corpus

A descriptive audit of the systematic extraction (Tables 4, 5, 6 and 7) shows that the reviewed PTR corpus contains a larger number of content-based studies than other paradigms, with hybrid systems forming the next largest group; collaborative filtering and purely knowledge-based systems appear less frequently, as summarized in Table 8. Reporting completeness varies across paradigms. Under the inclusive definition used in Table 8, most studies provide at least one quantitative evaluation indicator, but a subset does not report a standard predictive metric (e.g., Acc/AUC/F1) and instead emphasizes feasibility, clinician concordance, rule-consistency, or implementation-oriented outcomes, which limits direct metric-to-metric comparison across papers. Dataset reporting is similarly uneven: many studies explicitly state cohort size and provenance, while others provide only partial descriptors, constraining portability assessment. Table 8 summarizes these corpus-level counts and within-paradigm reporting rates computed directly from (Tables 4, 5, 6 and 7), grounding the cross-domain discussion in verifiable summary statistics.

3.5.2 Content-based methods

Content-based PTR models are most frequently applied in domains where patient state can be represented using relatively stable and clinically meaningful features (e.g., structured EHR covariates, laboratory values, or imaging-derived descriptors). (Tables 4, 5, 6 and 7) shows that these studies commonly report conventional predictive metrics on retrospective cohorts or public benchmarks, which supports within-study evaluation but does not guarantee cross-setting transportability. Their generalizability is sensitive to feature completeness, missingness patterns, and temporal drift; when records are sparse, rapidly evolving, or weakly structured, content-based approaches often require

Table 8 Descriptive summary of the extracted PTR corpus (computed from Tables 4, 5, 6 and 7; total $n = 60$)

Paradigm	#	Share of corpus (%)	Quantitative indicator reported (count; % within paradigm)	Dataset size reported (count; % within paradigm)
Content-based (CB)	38	63.3	37; 97.4	27; 71.1
Collaborative filtering (CF)	8	13.3	7; 87.5	6; 75.0
Knowledge-based (KB)	3	5.0	3; 100.0	0; 0.0
Hybrid	11	18.3	7; 63.6	6; 54.5

Share of corpus is computed over all included studies. *Quantitative indicator reported* denotes whether the paper reports at least one numeric evaluation result (including predictive metrics such as Acc/AUC/F1; ranking metrics such as nDCG/RMSE; concordance/coverage; or survival/outcome rates). *Dataset size reported* denotes whether an explicit cohort/sample size is stated. For the last two columns, values are shown as *count; percent within paradigm*

stronger representation learning, careful handling of confounding, and explicit robustness checks.

3.5.3 Collaborative filtering methods

Collaborative filtering (CF) approaches are typically positioned for settings with sufficient interaction history (e.g., repeated treatment–response observations, longitudinal prescribing/outcome logs, or large user–item matrices). In (Tables 4, 5, 6 and 7), CF studies most often operationalize effectiveness through prediction/ranking accuracy or reinforcement learning rewards under retrospective evaluation. These approaches can exploit population-level similarity to uncover latent therapeutic patterns, but are exposed to operational limitations that recur across domains: cold-start for new patients/therapies, susceptibility to sampling bias, and reduced stability in rare-disease or low-volume settings where interaction matrices are sparse or non-representative.

3.5.4 Hybrid methods

Hybrid PTR models are commonly motivated by heterogeneous inputs (e.g., EHR + labs + imaging + guideline constraints) and by clinical tasks where a single signal source is insufficient. Importantly, the extracted evidence supports a narrower, empirically grounded claim than "hybrids outperform": (Tables 4, 5, 6 and 7)indicates that hybridization is most frequently proposed to mitigate sparsity, fuse complementary modalities, or inject clinical constraints, and many hybrid studies report improvements over at least one stated baseline within their own experimental setup. Across the corpus, however, the credibility and interpretability of reported gains depend strongly on fusion design, the strength of baselines, and validation depth; hybrids without ablation/sensitivity analysis or with weak comparators can make attribution of benefit unclear and may increase complexity without commensurate clinical utility.

3.5.5 Knowledge-based methods

Knowledge-based PTR systems remain valuable in safety-critical or data-scarce contexts where explicit constraints from guidelines, expert rules, or ontologies are required to ensure plausibility and reduce unsafe recommendations. (Tables 4, 5, 6 and 7) suggests these systems are often evaluated using rule-consistency, concordance, or constrained recommendation objectives rather than predictive metrics. Their principal limitation is adaptability: purely knowledge-driven systems may struggle with evolving pathways or high inter-patient variability unless complemented by learning-based components.

3.5.6 Cross-cutting determinants of deployable performance

Across paradigms, (Tables 4, 5, 6 and 7) highlights recurring factors associated with stronger evidence of deployable effectiveness: (i) access to longitudinal and/or multimodal data with clear provenance, (ii) validation designs that extend beyond single-split retrospective testing (e.g., temporally separated evaluation, external cohorts, or multisite testing when reported), and (iii) outputs that are clinically interpretable and aligned with explicit decision points. Conversely, the most common barriers to translation are limited external validation, incomplete reporting of cohort/outcome definitions, and insufficient attention to usability, monitoring, and safety constraints. These cross-cutting

patterns motivate the trade-offs between model complexity, predictive performance, and clinical interpretability discussed in the following subsection.

3.6 Trade-offs between model complexity, predictive performance, and clinical interpretability

A recurring theme across the reviewed personalized treatment recommendation (PTR) literature is the three-way tension between *model complexity*, *predictive performance*, and *clinical interpretability*. Importantly, reported gains in conventional metrics (e.g., accuracy, AUC, F1) do not translate directly into clinical value unless the system can be inspected, validated, and operationalized within real decision pathways. Consequently, the practical objective is not to maximize complexity, but to select the *lowest-complexity* approach that achieves adequate performance while preserving transparency, safety, and workflow compatibility.

Complex architectures (e.g., deep neural models, sequential decision frameworks, and composite/hybrid designs) are typically motivated by settings with rich, heterogeneous signals and temporally evolving patient trajectories. In such contexts, additional representational capacity can help capture non-linear feature interactions, long-range dependencies, and multi-source evidence aggregation. However, these benefits are often paired with higher implementation burden: increased computational requirements, sensitivity to dataset shift, more design degrees of freedom that complicate ablation and error analysis, and a heavier validation requirement to establish reliability beyond a single retrospective cohort. From a deployment standpoint, opaque decision logic can further reduce clinician confidence unless the model exposes clinically meaningful rationales.

By contrast, lower-complexity approaches (including transparent feature-based models, guideline-constrained scoring logic, and conventional machine-learning pipelines) may underfit highly non-linear relationships, yet they offer tangible advantages for clinical translation. Their parameterization is easier to audit, failure modes are often easier to characterize, and validation protocols can be communicated more clearly to clinical and regulatory stakeholders. These properties become decisive in safety-sensitive or resource-constrained environments, and in applications where clinicians need to understand *why* a recommendation was produced to judge appropriateness for an individual patient.

Across the corpus, interpretability is most credible when it is treated as a first-class design requirement rather than an afterthought. Clinically usable explanations tend to be those that (i) map model outputs to recognizable decision points (e.g., treatment selection, dose adjustment, escalation), (ii) provide evidence-linked feature contributions or rule traces, and (iii) remain stable under reasonable perturbations of inputs. Post-hoc explanations can be informative, but they should be framed as *explanatory aids* unless accompanied by robustness checks that demonstrate consistency. Likewise, hybrid designs are best positioned when the integration strategy is explicit (e.g., structured fusion of data-driven predictions with guideline constraints) and when the added complexity is justified by improved generalization evidence or operational feasibility rather than marginal metric improvements alone.

Overall, the reviewed evidence supports a context-dependent selection principle: PTR systems should align algorithmic complexity with the clinical risk profile, data availability, and validation strength, while prioritizing explanation mechanisms that are

actionable for clinicians. In this sense, interpretability is not merely a usability feature; it is a prerequisite for trustworthy evaluation, accountable deployment, and sustained adoption in clinical workflows.

4 Real-world deployment and security challenges in ptr systems

PTR systems handle highly sensitive patient information and present distinct challenges related to privacy and data security. Since these systems depend on detailed personal details and extensive data exchanges, ensuring confidentiality, anonymity, and user control is essential to prevent breaches and maintain patient trust. This section highlights the critical privacy and security issues in PTR systems, emphasizing the need for robust protection to ensure safe and ethical personalized healthcare.

The main privacy concerns include the following:

- *Data Sensitivity and Confidentiality:* Health information, such as medical histories and genetic data, is extremely sensitive. Unauthorized access can lead to serious breaches of privacy and expose patients to discrimination and stigma.
- *Data Collection and Sharing:* To offer customized recommendations, a huge amount of patient data is often gathered, stored, and shared across various healthcare providers and systems. This widespread data sharing increases the likelihood of privacy violations, especially if third parties misuse data.
- *Anonymity and De-identification Challenges:* Even with data anonymization, advanced re-identification techniques can connect unidentified data with individual patients. Achieving true anonymity remains challenging, particularly when integrating data from multiple heterogeneous sources.
- *User Consent and Control:* Patients may lack an understanding of how their data are utilized in personalized treatment systems, raising ethical concerns about informed consent. Additionally, they often lack control over their data once they enter these systems, limiting their ability to withdraw consent or manage their data-sharing.

4.1 Challenges in data security for personalized treatment recommendation systems

Ensuring the confidentiality, integrity, and availability of patient data is a fundamental requirement for the safe deployment of personalized treatment recommendation systems (PTRS). Given the highly sensitive nature of medical records, genomic information, and behavioral data, PTRS are particularly vulnerable to security threats that can undermine patient trust and compromise clinical decision-making. The major data security challenges associated with PTR systems are summarized below:

- *Data Breaches and Unauthorized Disclosure:* Healthcare infrastructures are frequent targets of cyberattacks, and breaches can expose large volumes of sensitive patient information, including medical histories and personal identifiers. Such incidents pose severe risks to patient privacy and may result in regulatory penalties, financial losses, and long-term reputational damage for healthcare organizations.
- *Integrity and Authenticity of Recommendations:* Unauthorized manipulation of patient data, model parameters, or recommendation outputs can lead to incorrect or harmful treatment suggestions. Maintaining the integrity of both data pipelines and decision algorithms is therefore critical to ensuring the safety, reliability, and clinical validity of personalized recommendations.

- *Weak Authentication and Access Control Mechanisms:* In complex healthcare ecosystems involving multiple clinicians, institutions, and third-party service providers, inadequate authentication and access control mechanisms can enable unauthorized access to sensitive patient records. Such vulnerabilities increase the risk of data misuse and compromise compliance with healthcare privacy regulations.

4.2 Privacy-preserving and security-oriented solutions

To mitigate the aforementioned security risks, a range of technical and organizational strategies have been proposed and increasingly adopted within personalized treatment recommendation systems. These approaches aim to balance data protection with the need for effective model training and clinical utility:

- *Data Encryption and Secure Communication:* Strong cryptographic techniques are employed to protect patient data both at rest and during transmission. Encryption ensures that intercepted data remain unintelligible without appropriate decryption keys, thereby reducing the risk of unauthorized disclosure.
- *Access Control and Authentication Mechanisms:* The use of role-based access control, multi-factor authentication, and fine-grained permission models limits data access to authorized users only. These mechanisms are essential for maintaining accountability and preventing insider threats in multi-stakeholder healthcare environments.
- *Federated Learning:* Federated learning enables collaborative model training across distributed healthcare institutions without requiring the centralization of patient data. By keeping sensitive data locally and sharing only model updates, this paradigm significantly reduces exposure risks while supporting large-scale learning.
- *Differential Privacy:* Differential privacy techniques introduce controlled statistical noise into data or learning processes to prevent the re-identification of individual patients. This approach provides formal privacy guarantees while preserving the overall utility of analytical results.
- *Blockchain-Based Data Management:* Blockchain technology is increasingly explored for secure and transparent healthcare data management. Its decentralized and immutable ledger structure supports data integrity, auditability, and patient-centric control over data access, making it a promising solution for trust-sensitive medical applications.

4.3 Impact of privacy and security on patients

Privacy and security safeguards in personalized treatment recommendation systems (PTRS) play a pivotal role in shaping patient perceptions, trust, and long-term engagement with AI-driven healthcare technologies. Beyond technical protection, these measures directly influence patients' willingness to share sensitive health information and participate in data-driven care pathways.

- *Trust, Acceptance, and System Adoption:* Growing public awareness of data misuse and cyber threats has heightened patient sensitivity toward privacy risks. When PTRS demonstrate robust security guarantees and transparent data-handling practices, patients are more likely to trust these systems and actively engage with personalized treatment recommendations. Trust-driven adoption is particularly

critical in longitudinal care settings, where sustained patient participation is essential for system effectiveness.

- *Improved Recommendation Quality through Reliable Data Sharing:* Strong privacy assurances encourage patients to disclose accurate, complete, and context-rich health information. This, in turn, enhances the quality of patient profiles and enables PTRS to generate more precise and clinically relevant recommendations. Conversely, privacy concerns may lead to data withholding or distortion, negatively affecting recommendation accuracy and treatment outcomes.
- *Reduction of Psychological and Social Harm:* Privacy breaches or perceived risks of data exposure can induce psychological distress, including anxiety related to stigmatization, discrimination, or unauthorized secondary use of medical data. Effective security mechanisms mitigate these risks, contributing to a sense of safety and reinforcing patient confidence in AI-assisted clinical decision support systems.

4.4 Real-world deployment barriers

Despite substantial progress in algorithmic design and experimental validation, the translation of personalized treatment recommendation systems (PTRS) from research prototypes to real-world clinical environments remains challenging. Several structural, technical, and regulatory barriers constrain large-scale adoption and sustained clinical use:

- *Integration with Clinical Workflows and Health Information Systems:* Many PTRS are developed independently of existing hospital infrastructures, limiting their compatibility with electronic health records (EHRs) and clinical decision support systems. The absence of standardized interoperability frameworks, inconsistent data formats, and proprietary software ecosystems significantly increases deployment complexity and clinician workload.
- *Data Fragmentation and Heterogeneity:* Real-world healthcare data are distributed across multiple platforms, often characterized by missing values, inconsistent labeling, and heterogeneous acquisition protocols. Models trained on curated or homogeneous datasets may therefore experience performance degradation when deployed across diverse institutions, patient populations, and care settings.
- *Model Interpretability and Clinical Trust:* Although complex machine learning and deep learning models often achieve superior predictive performance, their lack of transparency limits clinical acceptance. Clinicians require interpretable explanations to validate recommendations, assess risk, and retain accountability for treatment decisions. The integration of Explainable AI (XAI) techniques is essential to bridge this trust gap and facilitate informed clinical adoption.
- *Computational and Infrastructure Constraints:* PTRS frequently demand substantial computational resources for training and real-time inference. Resource-limited environments, including rural healthcare facilities and mobile health units, may lack the infrastructure required to support these systems, restricting equitable access to AI-enabled personalized care.
- *Regulatory and Compliance Challenges:* Deployment in real-world settings must comply with stringent data protection and medical device regulations, such as HIPAA, GDPR, and region-specific clinical governance frameworks. In many jurisdictions, AI-based treatment recommendation tools require regulatory

approval from bodies such as the FDA or EMA, introducing additional validation requirements and prolonging deployment timelines.

4.5 Model interpretability and clinical trust

Model interpretability is a critical requirement for the successful adoption of personalized treatment recommendation systems (PTRS) in clinical practice. While advanced machine learning and deep learning models often demonstrate superior predictive performance, their opaque decision-making processes pose significant challenges in safety-critical healthcare environments. Clinicians are required to understand, justify, and take responsibility for treatment decisions, making transparency and explainability indispensable for clinical trust.

A substantial portion of existing PTRS relies on complex architectures such as deep neural networks, ensemble methods, or reinforcement learning models that function as "black boxes." Although these approaches can capture nonlinear and high-dimensional relationships within patient data, they provide limited insight into the factors driving specific treatment recommendations. This lack of interpretability not only hinders clinician confidence but also complicates error analysis, bias detection, and regulatory approval.

To address these challenges, Explainable Artificial Intelligence (XAI) techniques have emerged as essential components of trustworthy PTRS. Methods such as feature attribution, rule extraction, attention visualization, and surrogate modeling enable clinicians to interpret model outputs and assess the clinical plausibility of recommendations. Integrating interpretability mechanisms allows healthcare professionals to validate recommendations against medical knowledge, improving accountability and supporting shared decision-making with patients.

Despite these advancements, interpretability remains unevenly addressed across studies, with many systems prioritizing predictive accuracy over transparency. Achieving an effective balance between model complexity and explainability is therefore a key research challenge. Future PTRS must incorporate interpretability as a core design principle rather than an auxiliary feature to ensure safe, ethical, and clinically acceptable deployment.

4.6 Bias, fairness, and ethical considerations

Bias and fairness are central ethical challenges for personalized treatment recommendation systems (PTRS) because these models can reproduce (and sometimes amplify) structural inequities embedded in clinical data. Sources of bias observed across the reviewed studies include demographic imbalance, socioeconomic skew, site-specific practice patterns, and uneven data completeness, all of which can shift recommended therapies and downstream benefit toward well-represented groups. When PTRS are evaluated primarily with aggregate metrics, such effects may remain hidden, creating a risk of unequal clinical utility across subpopulations.

4.6.1 Geographic bias and representativeness (corpus-grounded audit)

In addition to demographic and institutional biases, *geographic bias* is a recurring concern because many PTRS are developed in health systems with mature digitization and standardized documentation. To ground this discussion in the reviewed corpus (rather

than inference alone), we conducted an explicit provenance audit using the dataset/source descriptors extracted in the systematic extraction tables (Tables 4, 5, 6 and 7). We coded a study's *dataset geography* only when the primary paper explicitly reported an identifiable country/region, institution, or national data source (e.g., named health system, registry, or cohort). If provenance was not stated (or only implicit via author affiliation), the study was conservatively coded as *not reportable* for geography.

Across the full set of 60 studies, dataset geography was explicitly identifiable for 18 studies based on information reported in the primary papers (e.g., named country/region, institution, registry, or national data source). Within this identifiable subset, 12/18 (66.7%) relied on Western datasets, 5/18 (27.8%) used non-Western datasets (predominantly East Asian health-system contexts), and 1/18 (5.6%) was conducted in an explicit LMIC setting. Stated conservatively at the corpus level, this yields a confirmed minimum of 12/60 (20.0%) Western-dataset studies and 1/60 (1.7%) LMIC-dataset studies. The true proportions may be higher because 42/60 studies did not report provenance with sufficient clarity to support reliable coding. This under-reporting is itself a portability-relevant transparency gap, because geographic and care-setting context is necessary for interpreting external validity and anticipating dataset-shift risks.

4.6.2 Evidence gap on geographic transfer

Crucially, within the reviewed literature we did *not* identify studies that explicitly evaluate *geographic transfer* (e.g., training in one region/health system and testing in another with a reported performance delta attributable to cross-country or cross-site shifts). Therefore, concerns about reduced performance or inappropriate recommendations under cross-geography deployment should be interpreted as theoretically motivated and consistent with known dataset-shift mechanisms, but not empirically demonstrated as transfer degradation within the current PTR corpus. Where non-Western data were available, they typically originated from single health systems or national datasets (e.g., South Korean EHR/insurance-linked settings), which may limit generalization beyond the originating care context [43, 55]. Likewise, commonly used Western benchmarks reflect region-specific documentation practices, formularies, and workflow constraints that may not transport without local recalibration and validation [11, 12, 38, 64].

4.6.3 Mitigation and responsible deployment

Mitigating bias in PTRS requires coordinated technical and governance measures. From a modeling and evaluation standpoint, subgroup-aware reporting (performance stratified by clinically relevant demographics), robustness checks under distribution shift, and fairness-aware objectives where appropriate are necessary to detect and reduce disparate impact. At a minimum, studies should report dataset provenance (country/region, care setting, inclusion period) and population descriptors to enable external scrutiny and replication. For deployment in data-scarce settings, incorporating knowledge-based constraints (e.g., guideline or safety rules) and clinician-in-the-loop oversight can reduce unsafe recommendations when purely data-driven generalization is uncertain. Ethically, PTRS should align with autonomy, beneficence, non-maleficence, and justice: patients require transparency about algorithmic involvement, clinicians require auditability and monitoring hooks, and systems should be validated in the intended care context before claims of broad generalizability are made.

5 Conclusion and future directions

This survey examined the evolving landscape of Personalized Treatment Recommendation Systems (PTRS) and their growing role in enabling patient-centered and data-driven healthcare. By systematically analyzing content-based, collaborative, knowledge-based, and hybrid recommendation paradigms, the review highlights how advances in artificial intelligence particularly deep learning, reinforcement learning, and hybrid modeling strategies have significantly enhanced the ability of PTRS to tailor therapeutic decisions to individual patient profiles. The predominance of applications in chronic disease management, including cardiovascular disorders and diabetes, reflects both pressing clinical needs and the suitability of PTRS for long-term, adaptive care.

Despite these methodological advances, the survey identifies persistent barriers that limit the translation of PTRS from experimental research to routine clinical practice. Key challenges include limited model interpretability, insufficient external validation, scalability constraints across heterogeneous healthcare environments, and unresolved concerns related to data privacy, fairness, and ethical compliance. Addressing these issues is essential for building clinician trust and ensuring responsible adoption of AI-driven decision support systems.

To support a more rigorous and clinically grounded evaluation of existing approaches, this study introduced a structured benchmarking framework that assesses PTRS across multiple dimensions, including algorithmic rigor, clinical applicability, validation depth, transparency, and ethical considerations. By shifting the focus from isolated performance metrics to a holistic assessment of methodological quality and translational readiness, the proposed rubric provides a systematic basis for comparative analysis across heterogeneous studies.

Looking ahead, future research should prioritize the development of interpretable and privacy-preserving PTRS that integrate seamlessly with clinical workflows and electronic health record systems. Greater emphasis is needed on standardized benchmarking practices, prospective and multi-institutional validation, and bias-aware model design to ensure equitable performance across diverse patient populations. Additionally, the incorporation of human-in-the-loop frameworks and regulatory-aligned evaluation protocols will be critical for bridging the gap between algorithmic innovation and real-world deployment. Collectively, these directions point toward the next generation of PTRS that are not only technically robust but also trustworthy, scalable, and clinically actionable in dynamic healthcare settings.

Appendix A Rubric-based assessment of reviewed studies

This appendix reports the rubric-based assessment used to summarize *evidence strength* across the reviewed studies in a transparent and auditable manner. The rubric is used as a structured synthesis framework (not a psychometric instrument) to standardize how methodological reporting, validation rigor, clinical actionability, and translational readiness are documented under heterogeneous study designs and data regimes.

Rubric application and auditability safeguards

Each included study was scored using predefined anchor criteria (Section 3.4 and Appendix A). Two authors independently evaluated the same set of papers, blinded to each other's initial ratings, and recorded brief per-dimension justification notes to maintain a traceable link between (i) what is explicitly reported in the source paper and (ii) the assigned score for each rubric dimension. This documentation step is intended to reduce ungrounded interpretation and to make scoring decisions inspectable by readers. To further support auditability at the table level, Table 9 includes an *Anchors* field that flags which key evidence items (dataset/source named, cohort size stated, evaluation design described, and quantitative indicator reported) were explicitly identifiable in each primary paper.

After the independent scoring pass, discrepancies were resolved using a structured adjudication protocol designed to minimize subjective negotiation: (i) re-check the source paper for explicit evidence relevant to the disputed dimension, (ii) re-apply the anchor criteria, (iii) when reporting was ambiguous or incomplete, apply a conservative interpretation rule by assigning the lowest score consistent with the available evidence, and (iv) if disagreement persisted, a third adjudicator reviewed the paper and issued the final decision with a documented rationale.

Scope note (replication and measurement claims)

The rubric is presented as an *auditability-oriented synthesis aid* rather than a psychometric measurement instrument. Accordingly, we do not present formal rater-agreement evidence for independent replication in this revision (e.g., percent agreement, weighted κ, ICC). Reconciled scores are therefore used only for descriptive synthesis and for organizing discussion, bounded by what is explicitly reported in each primary study.

Worked scoring examples with evidence-linked justifications are provided in Table 10 to make the scoring logic auditable in practice. For studies not shown in Table 10, auditability is supported via the corpus-wide *Anchors* field in Table 9 together with the study-level extraction Tables 4–7, which record the reportable dataset descriptors, evaluation descriptors, and quantitative indicators used to ground scoring.

Evidence-strength strata (descriptive groupings)

For descriptive summarization, we optionally report *evidence-strength strata* derived from the aggregate score (mean across the five scored dimensions). These strata are intended to highlight recurring *reporting and model-evaluation patterns* (e.g., limited documentation, single-cohort retrospective testing, external validation, deployment-oriented considerations) rather than to render definitive judgments of scientific merit or paradigm superiority. Thresholds and their rationale are specified in Table 9 footnote and applied consistently across the reviewed corpus.

Operationalization of the rubric and scoring criteria

To support transparent cross-study synthesis under substantial heterogeneity in objectives, datasets, and evaluation protocols, we operationalized the rubric as five scored dimensions on a 10-point ordinal scale (1–10), where higher values indicate stronger *evidence strength* with respect to methodological rigor, clinical actionability, and translational readiness. The rubric is used as a structured synthesis aid, not as a psychometric

instrument for definitive quality ranking. The *recommendation paradigm* (content-based, collaborative filtering, knowledge-based, hybrid) is recorded as a descriptive taxonomy field for stratified analysis and is not included in the aggregate score.

Scores were assigned using predefined *anchor criteria* that map commonly observed, verifiable reporting patterns to score ranges. Anchors prioritize what is explicitly documented in each paper (e.g., validation design, data provenance/preprocessing description, clinically meaningful outputs, deployment-relevant considerations) to reduce interpretive drift. When information required by an anchor was not explicitly reported, we applied a conservative interpretation rule by assigning the lowest score consistent with the available evidence.

Algorithmic rigor (1–10)

Assesses methodological transparency and robustness, including clarity of model specification, justified design choices, baseline selection, and robustness checks. Lower scores (1–4) reflect insufficient methodological detail (e.g., unclear training/validation pipeline, missing baselines). Mid-range scores (5–7) indicate standard modeling with adequate description but limited robustness evidence (e.g., minimal sensitivity/ablation analysis or weak baseline justification). Higher scores (8–10) are reserved for studies providing strong methodological completeness and robustness support (e.g., well-justified baselines, sensitivity/ablation analysis, calibration or error analysis, and clearly motivated modeling choices).

Scoring procedure and auditability protocol

Each included study was scored using the anchor criteria defined above. Two authors independently rated the same set of papers, blinded to each other's initial scores, and recorded brief per-dimension justification notes that link scores to explicitly reported evidence in the primary paper (e.g., dataset/source description, evaluation design, clinical objective, and stated limitations). This procedure improves transparency and traceability of the synthesis.

After the independent scoring pass, discrepancies were resolved using a structured adjudication protocol intended to minimize subjective negotiation: (i) re-check the paper for explicit evidence relevant to the disputed dimension, (ii) re-apply the anchor criteria, (iii) when reporting was ambiguous or incomplete, apply a conservative rule by assigning the lowest score consistent with the available evidence, and (iv) if disagreement persisted, a third adjudicator reviewed the paper and issued a final decision with a documented rationale.

To make the scoring logic inspectable in practice, Table 10 provides worked examples with evidence-linked justifications for a representative subset of studies. Reconciled scores are used only for descriptive synthesis and trend analysis. The rubric is presented as an auditability-oriented synthesis aid, and rubric-derived strata are treated as descriptive evidence-strength groupings rather than stand-alone proof of paradigm superiority; accordingly, comparative claims about effectiveness are restricted to studies that report direct head-to-head evaluations under comparable validation conditions.

Table 9 Rubric-based *evidence-strength* assessment of personalized treatment recommendation studies

References	Year	Type	Source	Paradigm	Alg.	Clin.	Valid.	Transp.	Scale/ Expl.	Mean	Stratum	Anchors
[6]	2023	Conf.	IEEE	CB	9	8	9	8	8	8.4	High	D N V M
[7]	2021	Conf.	IEEE	CB	8	8	7	7	7	7.4	Moderate	D N V M
[8]	2021	Conf.	IEEE	KB	7	6	6	6	6	6.2	Moderate	D V M
[9]	2024	Journal	IEEE	CB	7	7	6	6	6	6.4	Moderate	D N V M
[10]	2022	Conf.	IEEE	Hybrid (CB+KB)	8	8	7	7	8	7.6	Moderate	D N V M
[11]	2023	Conf.	IEEE	CF	9	8	9	8	8	8.4	High	D N V M
[12]	2024	Conf.	IEEE	CB	8	8	8	7	7	7.6	Moderate	D V M
[13]	2020	Conf.	IEEE	CB	7	6	7	6	6	6.4	Moderate	D N V M
[14]	2022	Conf.	IEEE	Hybrid (CB+CF)	8	8	8	8	8	8.0	High	D V M
[15]	2023	Journal	Springer	Hybrid (CB+KB)	6	5	5	5	5	5.2	Moderate	D V M
[36]	2024	Journal	IEEE	CF	9	9	9	9	9	9.0	High	D V M
[16]	2023	Conf.	IEEE	CF	8	8	8	7	8	7.8	Moderate	D N V M
[17]	2024	Conf.	IEEE	CB	6	6	5	5	5	5.4	Moderate	D V M
[18]	2020	Journal	IEEE	KB	8	9	9	8	9	8.6	High	D V M
[19]	2023	Conf.	IEEE	CB	7	7	6	6	6	6.4	Moderate	D V
[20]	2023	Conf.	IEEE	CB	9	9	9	8	8	8.6	High	D N V M
[21]	2024	Journal	Springer	CB	8	9	8	8	7	8.0	High	D V
[22]	2021	Journal	Springer	CF	7	8	7	7	7	7.2	Moderate	D N V M
[23]	2024	Journal	Springer Nature	KB	7	8	6	6	6	6.6	Moderate	D V
[24]	2024	Journal	Springer	CB	6	8	6	6	6	6.4	Moderate	D N V
[25]	2023	Journal	MDPI	KB	9	9	8	8	7	8.2	High	D N V M
[26]	2023	Journal	Springer	CB	8	8	7	7	7	7.4	Moderate	D N V M
[71]	2023	Journal	Science Direct	KB	8	9	8	8	7	8.0	High	D N V M
[55]	2021	Journal	Taylor & Francis	Hybrid (CB+Statistical Learning)	8	8	7	7	7	7.4	Moderate	D N V M
[32]	2023	Journal	Wiley	CB	8	8	8	7	7	7.6	Moderate	D N V M
[27]	2023	Journal	Springer Nature	CF	8	9	8	7	8	8.0	High	D N V M
[33]	2023	Journal	Springer Nature	CB	9	9	9	8	8	8.6	High	D N V M
[34]	2023	Journal	Springer Nature	KB	7	8	6	6	7	6.8	Moderate	D N V M

Table 9 (continued)

References	Year	Type	Source	Paradigm	Alg.	Clin.	Valid.	Transp.	Scale/ Expl.	Mean	Stratum	Anchors
[35]	2022	Journal	Science Direct	KB	9	9	8	8	8	8.4	High	D N V M
[37]	2023	Journal	Springer Nature	KB	7	7	6	6	6	6.4	Moderate	D V
[38]	2023	Conf.	IEEE	Hybrid (CB+KB)	9	9	8	9	9	8.8	High	D N V M
[39]	2023	Conf.	IEEE	CB	8	8	7	8	8	7.8	Moderate	D N V M
[43]	2023	Journal	Science Direct	KB	8	8	8	7	8	7.8	Moderate	D V
[44]	2024	Journal	Elsevier	Hybrid (CB+KB)	9	9	9	9	9	9.0	High	D N V M
[45]	2023	Journal	Elsevier	Hybrid (CB+KB)	9	8	9	9	9	8.8	High	D N V M
[46]	2022	Journal	Elsevier	CB	7	7	7	8	7	7.2	Moderate	D N V M
[47]	2024	Journal	Science Direct	CB	5	5	5	6	5	5.2	Moderate	D V
[48]	2019	Journal	MDPI	CF	6	6	5	6	6	5.8	Moderate	D V
[49]	2022	Journal	IEEE	CB	8	7	8	8	8	7.8	Moderate	D V M
[53]	2022	Journal	Springer	CB	6	7	6	6	6	6.2	Moderate	D V M
[54]	2022	Journal	Wiley	CF	8	9	8	9	9	8.6	High	D N V M
[60]	2024	Conf.	IEEE	CB	7	7	6	6	7	6.6	Moderate	D V
[40]	2021	Conf.	IEEE	CB	8	9	8	9	9	8.6	High	D N V M
[61]	2022	Journal	BMC	Hybrid (CB+KB)	9	8	9	9	9	8.8	High	D N V M
[62]	2023	Journal	Scopus	CB	6	6	6	5	6	5.8	Moderate	D V
[63]	2023	Journal	MDPI	CB	8	7	8	7	8	7.6	Moderate	D N V M
[64]	2023	Journal	Science Direct	CB	7	7	7	7	7	7.0	Moderate	D N V M
[65]	2019	Journal	Science Direct	CB	7	6	7	7	7	6.8	Moderate	D N V M
[66]	2023	Journal	Elsevier	CB	8	8	7	8	8	7.8	Moderate	D V
[67]	2023	Journal	Elsevier	CB	9	8	8	9	9	8.6	High	D N V M
[68]	2024	Journal	Elsevier	Hybrid (CB+KB)	9	9	9	9	9	9.0	High	D N V M
[69]	2024	Journal	Elsevier	CB	8	9	8	8	8	8.2	High	D N V M

Table 9 (continued)

References	Year	Type	Source	Paradigm	Alg.	Clin.	Valid.	Transp.	Scale/ Expl.	Mean	Stratum	Anchors
[70]	2019	Journal	MDPI	Hybrid (CB+KB)	8	7	8	7	8	7.6	Moderate	D N V M
[4]	2025	Journal	Elsevier	CB	8	9	8	9	9	8.6	High	D N V M

"Mean" denotes the arithmetic mean of the five scored dimensions (Alg., Clin., Valid., Transp., Scale/Expl.)

"Paradigm" is descriptive and is not scored. "Mean" denotes the arithmetic mean of Alg., Clin., Valid., Transp., and Scale/Expl. "Stratum" denotes an evidence-strength grouping used for descriptive synthesis rather than a definitive quality rank; thresholds (High ≥ 8.0, Moderate 5.0–7.9, Lower < 5.0) were applied consistently across the corpus. *Anchors* flags which evidence items were explicitly identifiable in the primary paper and therefore available to substantiate scoring: D = dataset/source named; N = cohort/sample size stated; V = evaluation/validation design described; M = at least one quantitative indicator reported. Worked, evidence-linked scoring examples are provided in Table A2 to make the scoring rationale inspectable in practice

Appendix B Algorithmic workflows for PTR paradigms

This appendix consolidates the algorithm pseudocodes referenced throughout the main text and presents them as *workflow templates* for the major personalized treatment recommendation (PTR) paradigms reviewed in this survey. These pseudocodes are provided to improve *traceability* of the pipeline logic (inputs, intermediate representations, and decision outputs) and to offer a consistent reference point when interpreting comparative findings across heterogeneous studies. They are *not* intended as executable implementations, nor as prescriptive system designs; instead, they abstract common processing steps observed across the literature.

Algorithm 1 Content-based filtering model for treatment recommendation

Require: Patient profile P, Treatment database T, Similarity function Sim()
Ensure: Recommended treatments
Extract patient features from P
Extract treatment features from T
Initialize an empty list R
for each treatment t_i in T **do**
    Compute similarity score $s_i = \text{Sim}(P, t_i)$
    Append (t_i, s_i) to R
end for
Sort R based on similarity scores in descending order
return Top-N treatments from R

Algorithm 2 Collaborative-filtering based treatment recommendation

Require: Matrix M, Patient P_t, Similarity Sim()
Ensure: Recommended Treatments for P_t
Extract history H_t of P_t from M
Initialize list S
for each P_i in M **do**
    $s_i \leftarrow \text{Sim}(P_t, P_i)$
    Append (P_i, s_i) to S
end for
Sort S by s_i in descending order
Select top-k similar patients
Initialize list T
for each treatment T_j **do**
    Compute score $\hat{s}_j$ via weighted CF
    Append $(T_j, \hat{s}_j)$ to T
end for
Sort T by $\hat{s}_j$ in descending order
return Top-N treatments from T

Table 10 Worked rubric scoring examples with evidence-linked justifications (illustrative subset; aligned with Table 9)

Study (example)	Alg.	Clin.	Valid.	Transp.	Scale/Expl.	Evidence-linked justification (dimension-wise)
[6]	9	8	9	8	8	*Alg:* EEG-based stress classification integrated with a recommendation component (SVM + K-means); performance reported on a public EEG dataset.*Clin:* targets a concrete therapeutic objective (stress alleviation) using a defined intervention modality (music suggestions); workflow integration not detailed.*Valid:* evaluation on a public dataset with stated experimental scale (34 experiments) and quantified accuracy (96.3%).*Transp:* dataset provenance and acquisition context (EMOTIV; public dataset) described; limitations explicitly stated.*Scale/Expl:* computationally feasible (standard ML); no explicit clinician-facing explanation mechanism reported.
[7]	8	8	7	7	7	*Alg:* Random Forest preference prediction with feature selection/processing; method described at a replicable level.*Clin:* tinnitus music-therapy personalization with non-repetitive recommendations mapped to a supportive-care decision.*Valid:* retrospective evaluation on 30 patients (450 sessions); limited sample size constrains generalization evidence.*Transp:* cohort size and data collection context reported; confounders (mood/sleep) discussed as limitations.*Scale/Expl:* scalable algorithms, but interpretability/deployment considerations not elaborated in the summary.
[8]	7	6	6	6	6	*Alg:* clinical-trial ranking formulation using Rocchio + LambdaRank; robustness/ablation evidence not described in the summary.*Clin:* framed for oncology trial matching but limited description of workflow/actionability beyond relevance improvement.*Valid:* benchmark-based evaluation on TREC 2017 using nDCG; indicates IR benchmarking rather than clinical validation.*Transp:* benchmark source stated; limited discussion of clinical data provenance/preprocessing in the provided summary.*Scale/Expl:* computational feasibility plausible; explicit clinical interpretability mechanisms not described.
[38]	9	9	8	9	9	*Alg:* reinforcement learning (Q-learning family) for sequential decision support; formulation suited to personalized treatment policies.*Clin:* emphasizes individualized treatment/adherence improvement and outcome optimization aligned with clinical decision processes.*Valid:* evaluated on EHR-derived data (South Korean EHRs); external/prospective testing not specified in the summary.*Transp:* data source type stated and ethical implications explicitly acknowledged as a gap.*Scale/Expl:* deployment-relevant considerations highlighted (ethics/safeguards); specific explainer (e.g., SHAP) not reported in the summary.

Table 10 (continued)

Study (example)	Alg.	Clin.	Valid.	Transp.	Scale/Expl.	Evidence-linked justification (dimension-wise)
[18]	8	9	9	8	9	*Alg:* ranking-based drug sensitivity prediction (Q-Rank) described as integrating and improving upon prior algorithms, supporting a strong methodological contribution. *Clin:* targets treatment selection in oncology (drug sensitivity), directly linked to therapeutic decision-making. *Valid:* evaluated on a named benchmark dataset (NCI-DREAM7) with comparative performance against baselines, supporting strong benchmarking evidence. *Transp:* benchmark dataset is identified and the limitation (dataset specificity/generalizability) is explicitly discussed. *Scale/Expl:* ranking outputs support clinical interpretability (ordered therapies) and are operationally feasible for decision support.
[54]	8	9	8	9	9	*Alg:* collaborative-filtering–style PTR (as categorized in Table 9) with high rubric scores suggests a well-specified modeling pipeline in the reviewed paper. *Clin:* positioned for clinically meaningful personalization (high clinical applicability score), indicating clear linkage to actionable care decisions. *Valid:* scored as strong retrospective/benchmark evidence (8), consistent with robust evaluation design reported in the primary study. *Transp:* high transparency score (9) indicates explicit data provenance and preprocessing documentation in the reviewed paper. *Scale/Expl:* high scale/explainability score (9) indicates explicit interpretability and/or deployment feasibility considerations reported.

Table 10 (continued)

Study (example)	Alg.	Clin.	Valid.	Transp.	Scale/Expl.	Evidence-linked justification (dimension-wise)
[36]	9	9	9	9	9	*Alg:* top-tier algorithmic rigor indicates a fully specified approach with strong design justification and robust benchmarking in the reviewed paper. *Clin:* highest clinical applicability reflects explicit alignment to clinical decision-making and actionable outputs. *Valid:* maximum validation depth indicates strong generalization evidence (e.g., rigorous benchmarking and/or external testing as reported in the primary study). *Transp:* maximum transparency/ethics reflects comprehensive reporting of data provenance and governance statements. *Scale/Expl:* maximum score indicates explicit explainability and feasibility considerations supporting adoption.
[44]	9	9	9	9	9	*Alg:* highest algorithmic rigor reflects a thoroughly specified hybrid PTR approach with robust methodological support in the reviewed paper. *Clin:* highest clinical applicability indicates clear mapping to clinical decision contexts. *Valid:* maximum validation depth indicates strong generalization/benchmarking evidence as reported. *Transp:* maximum transparency indicates comprehensive reporting of data and processing steps with governance/ethics attention. *Scale/Expl:* maximum score indicates explicit interpretability and deployment feasibility considerations.

The rightmost column links each score to concrete, verifiable attributes reported in the corresponding primary study, to support auditability of the scoring logic

Justifications are restricted to information explicitly stated in the corresponding primary paper. When a dimension is not documented (e.g., no named explainability method), we do not infer missing details; we instead record only what is reportable and apply the rubric conservatively. The goal is a traceable link from reported study characteristics (data provenance, validation design, clinical objective, stated limitations) to each assigned score

Algorithm 3 Knowledge-based filtering treatment recommendation

Require: Patient Profile P, Knowledge Base K, Rule-Based Engine R
Ensure: Recommended Treatments T
Extract patient-specific data D (e.g., medical history, symptoms, risk factors) from P
Retrieve clinical guidelines and expert knowledge from K
Initialize empty list S
for each candidate treatment t **do**
    Evaluate t using rule-based matching via R
    Compute relevance score r_t based on:

- Alignment with clinical guidelines
- Treatment efficacy from prior outcomes
- Patient-specific risk and contraindication profiles

    Append (t, r_t) to S
end for
Sort S by r_t in descending order
Select top-N treatments
return Recommended treatments T

Algorithm 4 Hybrid filtering treatment recommendation

Require: Patient profile P, Treatment database D, Similarity function Sim(), Knowledge base KB, Fusion method Fusion_Method
Ensure: Recommended Treatments T
Extract patient-specific features from P
Retrieve available treatments from D ▷ Step 1: Content-Based Filtering
for each treatment t in D **do**
    Compute similarity score $s_{CB}(t) = \text{Sim}(P, t)$
end for ▷ Step 2: Collaborative Filtering
Identify top-k similar patients
Predict treatment scores $s_{CF}(t)$ using collaborative filtering ▷ Step 3: Knowledge-Based Filtering
for each treatment t in D **do**
    Evaluate t using knowledge base KB and assign score $s_{KB}(t)$
end for ▷ Step 4: Hybrid Fusion
for each treatment t in D **do**
    Compute final score $s_{\text{final}}(t) = \text{Fusion_Method}(s_{CB}(t), s_{CF}(t), s_{KB}(t))$
end for ▷ Step 5: Ranking and Selection
Rank all treatments in D by $s_{\text{final}}(t)$ in descending order
return Top-N treatments

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Author contributions
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Data availability
No datasets were generated or analysed during the current study.

Code availability
Not applicable.

Materials availability
Not applicable.

Declarations

Ethics approval and consent to participate
Not applicable. This manuscript is a literature survey and does not involve human participants, animals, or any new data collection by the authors.

Clinical trial number
Not applicable. This study does not involve any clinical trials.

Consent for publication
All authors have reviewed and approved the final version of this manuscript and consent to its publication. Since the study does not include any identifiable individual data or participant information, additional consent for publication was not required.

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